

结直肠癌休眠细胞复苏的转录组线索

王瀚郭鹏程

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研究背景与核心假设

- 结直肠癌远处复发可能并非单纯由晚期转移扩散决定，部分播散或残留癌细胞可能在早期已进入远处生态位。
- 本课题关注休眠癌细胞从低增殖状态重新苏醒的分子触发因素，尤其是转录调控、通路活化与微环境信号。
- 核心假说：若能识别并阻断促使休眠细胞复苏的关键轴，就可能使残留癌细胞长期维持休眠，从而降低复发与转移风险。
- 本页汇报先以 GSE114012 的 LRC/BULK RNA-seq 模型建立可复现的候选基因证据链，为后续 TF、通路和药物重定位分析提供入口。

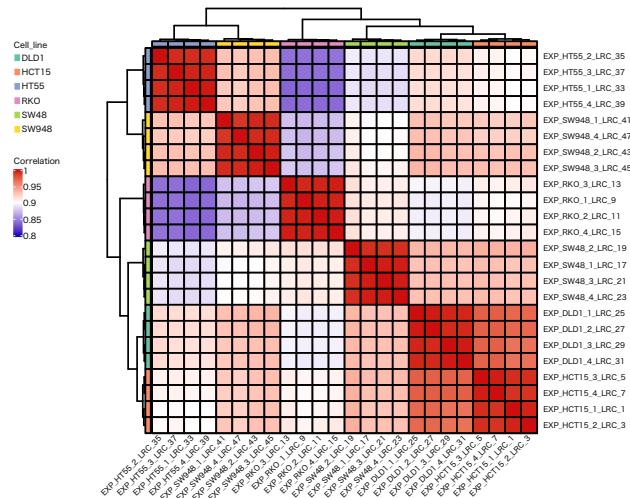
GSE114012 数据模型与本次分析任务

- GEO 数据集 GSE114012 的主题是识别结直肠癌球体培养中的休眠细胞；官方设计使用 CFSE 标记并分选保留染料的 LRC 与增殖/循环状态的 BULK 细胞。
- 数据包含 DLD1、HCT15、HT55、SW948、RKO、SW48 六种结直肠癌细胞系，每种细胞系包含 LRC 与 BULK 群体的重复样本。
- 本次分析将 LRC 视作休眠样细胞状态、BULK 视作相对循环/增殖状态，重点比较 LRC vs BULK 的表达差异与跨细胞系稳定交集。
- 所有差异分析、交集和火山图共享同一显著性阈值，以保证后续候选“休眠维持/复苏相关基因”的可比性。

脚本流程与结果约束

1. **00**: 使用 TPM 做 LRC 与 BULK 样本层级聚类，先检查休眠样/增殖样样本结构与离群情况。
2. **01**: 基于 limma-voom 进行 LRC vs BULK 差异分析，得到每个细胞系及组合设计的显著基因。
3. **02**: 按可配置细胞系组合提取显著基因交集，筛选跨模型重复出现的候选休眠相关信号。
4. **03**: 用传统火山图观察单个比较中的表达漂移、显著性强度和候选基因位置。
5. **04**: 用多组火山图并列比较多套细胞系组合，突出方向一致的上调/下调信号。
6. **05**: 提取每套差异分析设计的 Top 50 显著差异基因，并用 TPM 行 Z-Score 热图展示 LRC/BULK 表达模式。

00. LRC 样本热图 (TPM)



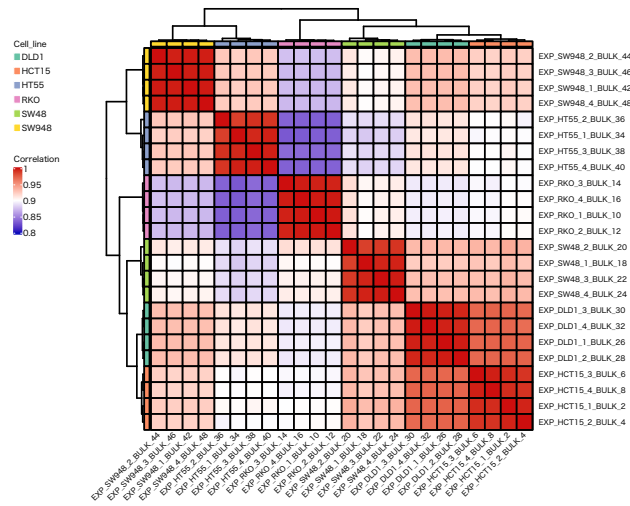
说明：

- LRC 是 CFSE 保留的休眠样群体，聚类用于观察不同细胞系休眠样样本是否呈现相近表达结构；
- 若 LRC 样本之间存在稳定聚类，说明休眠样状态可能具有跨模型共享的转录轮廓；
- 该图为后续差异分析提供样本质量与分组结构依据。

相对路径：

results/ngs/GSE114012/plots/sample_clustering_heatmap/coding/
LRC/heatmap.pdf

00. BULK 样本热图 (TPM)



说明：

- BULK 代表相对循环/增殖状态，用于与 LRC 休眠样群体建立背景对照；
- 聚类结构可帮助判断 BULK 样本是否主要受细胞来源或增殖状态驱动；
- 与 LRC 热图对照可辅助解释休眠样表达差异是否具有模型一致性。

相对路径：

results/ngs/GSE114012/plots/sample_clustering_heatmap/coding/
BULK/heatmap.pdf

01. 差异分析目录与统一输出结构

每个分组分析都输出如下固定三张表，放在统一目录：

`results/ngs/GSE114012/tables/<Analysis_Name>/DEG/`

- `all_genes.csv`
- `significant_genes.csv`
- `summary.csv`

下方所有“单张表页”按同一风格逐一展示，默认给出文件前若干行以便快速核对字段与结果质量。

01. 差异分析 Summary: ALL

说明：该表概括当前 LRC vs BULK 比较的样本数、显著上调/下调基因数和阈值设置，用于判断每个细胞系或组合模型中休眠样 LRC 相对 BULK 的总体转录变化强度。

相对路径： `results/ngs/GSE114012/tables/ALL/DEG/summary.csv`

表格预览：展示全部列与前 21 行。

Dataset	Data_Type	Analysis_Name	Contrast	Samples_Used	Up	Down	Total_Significant_Genes	P_Value_Column	P_Value_Cutoff	LogFC_Cutoff	Microarray_Log2_Transformed
GSE114012	ngs	ALL	ALL_vs_BULK	48	65	0	65	P.Value	0.05	0.5	-

01. ALL: all_genes

说明：该表保留完整差异分析排序结果，可用于追溯每个基因在 LRC/BULK 模型中的 logFC、显著性和排序位置，是后续候选 TF、通路和药物重定位输入的基础。

相对路径： results/ngs/GSE114012/tables/ALL/DEG/all_genes.csv

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000099783	HNRNPM	ENSG00000099783	4670	-0.315207173428172	7.72338094641144	-5.47673768020494	1.54358066486785e-06	0.0121685824817598	4.91002038968939
ENSG00000136450	SRSF1	ENSG00000136450	6426	-0.424998465000331	7.69661367030826	-5.42153481081821	1.87007568491775e-06	0.0121685824817598	4.74138388356755
ENSG00000099800	TIMM13	ENSG00000099800	26517	-0.290351583373309	5.37061805645465	-4.9982952554396	8.03274045454908e-06	0.0324847987244109	3.40330034810751
ENSG00000122884	P4HA1	ENSG00000122884	5033	0.437817019402286	6.59107001919331	4.91359129107831	1.07171231973372e-05	0.0324847987244109	3.20314280405155
ENSG00000129351	ILF3	ENSG00000129351	3609	-0.226621907746059	8.63290463159867	-4.86867995718228	1.248071258814e-05	0.0324847987244109	3.0543476825272
ENSG00000116560	SFPQ	ENSG00000116560	6421	-0.282165935258554	8.33201570202119	-4.74541270290884	1.89217483241049e-05	0.0410412721149836	2.69579069549393
ENSG00000071626	DAZAP1	ENSG00000071626	26528	-0.313842935262246	6.8353147662722	-4.59370754459055	3.14405656124182e-05	0.0584525029828587	2.2574045552503
ENSG00000282283	PLPPR3	ENSG00000282283	79948	-0.220308633353087	5.70682823362442	-4.32638136147107	7.59255427115289e-05	0.12351187660598	1.46610874897111
ENSG00000147274	RBMX	ENSG00000147274	27316	-0.218241644242318	7.59385078927674	-4.25964075711669	9.43468946284677e-05	0.130496018786412	1.29095432787921
ENSG00000011304	PTBP1	ENSG00000011304	5725	-0.1938302490346	7.96255699859407	-4.2408539812519	0.000100273565995399	0.130496018786412	1.23574650247937
ENSG00000182944	EWSR1	ENSG00000182944	2130	-0.300768497303325	7.79248819568433	-4.15547111774501	0.000132094413989924	0.140035498403357	0.994595988782982
ENSG00000161547	SRSF2	ENSG00000161547	6427	-0.344444065679467	7.25975726728635	-4.1540940239892	0.00013268059763164	0.140035498403357	0.991658866769967
ENSG00000122566	HNRNPA2B1	ENSG00000122566	3181	-0.326508380881387	9.47895777562502	-4.10227330802366	0.000156671608260439	0.140035498403357	0.831604544249604
ENSG00000175224	ATG13	ENSG00000175224	9776	0.151013422443921	6.71802019691976	4.06261974108176	0.000177821908710464	0.140035498403357	0.733953667248106
ENSG00000225553	PHF1	ENSG00000225553	5252	0.490022390805626	5.46329166487175	4.06138423581968	0.000178523493189313	0.140035498403357	0.716038030773765
ENSG00000170144	HNRNPA3	ENSG00000170144	220988	-0.32203693266831	7.88380810746506	-4.0511913543319	0.000184414820150343	0.140035498403357	0.701555181669092
ENSG00000197746	PSAP	ENSG00000197746	5660	0.328086100884395	9.87639354528802	4.04307110769361	0.000189242613631607	0.140035498403357	0.663573925500827
ENSG00000091483	FH	ENSG00000091483	2271	-0.25860797256903	6.11673489069131	-4.03577171619488	0.000193686719783343	0.140035498403357	0.655976462552801
ENSG00000284254	HNRNPH1	ENSG00000284254	3187	-0.178156854671239	8.08038071409568	-3.95805153624202	0.0002477291316213	0.156862995183924	0.442425258160037
ENSG00000135828	RNASEL	ENSG00000135828	6041	0.393143440908849	3.19239947527926	3.95580636111211	0.000249489274948045	0.156862995183924	0.0865227863261069
ENSG00000163395	IGFN1	ENSG00000163395	91156	1.36608487520866	-1.26922185228838	3.95122126636515	0.000253121476783648	0.156862995183924	-2.90877767823594

01. ALL: significant_genes

说明: 该表按统一阈值筛选显著差异基因, 代表当前模型中最直接的休眠样 LRC 转录特征, 并作为交集分析和火山图展示的核心输入。

相对路径: results/ngs/GSE114012/tables/ALL/DEG/significant_genes.csv

表格预览: 展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000163395	IGFN1	ENSG00000163395	91156	1.36608487520866	-1.26922185228838	3.95122126636515	0.000253121476783648	0.156862995183924	-2.90877767823594
ENSG00000105499	PLA2G4C	ENSG00000105499	8605	1.41365694646593	-0.94883288840517	3.73111874100223	0.000502320407617984	0.233471349455016	-2.92869879875184
ENSG00000162772	ATF3	ENSG00000162772	467	1.22685931850981	3.65107520268085	3.63517016802484	0.000673539249217337	0.257807052626895	-0.575905809763909
ENSG00000186517	ARHGAP30	ENSG00000186517	257106	0.783246799644915	0.199991090954644	3.16371670499563	0.00269593846956254	0.596830257975345	-2.91818618936256
ENSG00000158406	H4C8	ENSG00000158406	8365	0.96510849740796	-0.141094037106581	3.16243319480387	0.00270577725684227	0.596830257975345	-3.06504829672572
ENSG00000148926	ADM	ENSG00000148926	133	0.76950482087548	3.89815208796736	3.12320037661417	0.00302335425873032	0.63699923169218	-1.71934029794743
ENSG00000172602	RND1	ENSG00000172602	27289	0.780294414485237	0.871227051428515	3.04131719877687	0.00380239358073999	0.703513581839077	-2.77621690899782
ENSG00000184205	TSPYL2	ENSG00000184205	64061	0.798021448812101	4.69095580832635	2.99854096665734	0.00428065417289403	0.754645754900318	-1.9948572508155
ENSG00000126368	NR1D1	ENSG00000126368	9572	0.801768027569022	5.24264131370404	2.95697913019277	0.00479872486860272	0.81104682389605	-2.10280105958819
ENSG00000175197	DDIT3	ENSG00000175197	1649	0.833920926252041	4.71782671997339	2.93611964495266	0.00508025918345147	0.82668020777986	-2.13718104511576
ENSG00000156804	FBXO32	ENSG00000156804	114907	0.833981805223282	4.2464633204439	2.93406442405098	0.00510881572062549	0.82668020777986	-2.13417169072631
ENSG00000131480	AOC2	ENSG00000131480	314	1.03556627952865	0.836522726553695	2.91615970995255	0.00536399878447488	0.831036668823287	-2.9552938337427
ENSG00000173110	HSPA6	ENSG00000173110	3310	1.03572300608848	-0.233713912986225	2.88853465968441	0.00578109466630448	0.847868825131127	-3.3592530366125
ENSG00000223573	TINCR	ENSG00000223573	257000	1.63625545835788	-0.355792789665992	2.88742926063676	0.00579839599175275	0.847868825131127	-3.4176366098075
ENSG00000118523	CCN2	ENSG00000118523	1490	1.16913472761058	1.56260975062346	2.81959913387589	0.00695693242056974	0.871489542657836	-2.82610960363345
ENSG00000172361	CFAP53	ENSG00000172361	220136	0.708803925163841	0.336672789739038	2.8195452681653	0.00695793221702625	0.871489542657836	-3.2312980063991
ENSG00000240891	PLCXD2	ENSG00000240891	257068	0.64425351137578	0.379631847939463	2.81795753977273	0.00698746173148749	0.871489542657836	-3.21712275535942
ENSG00000118503	TNFAIP3	ENSG00000118503	7128	0.694882974917777	2.74805355204586	2.81695414322357	0.00700618329876119	0.871489542657836	-2.48525436292323
ENSG00000196611	MMP1	ENSG00000196611	4312	2.6128447212111	-0.835158541455007	2.81205045304794	0.00709834728152225	0.871489542657836	-3.65172934832994
ENSG00000174844	DNAH12	ENSG00000174844	201625	1.26186187339818	-0.848187158675002	2.7097962599604	0.00929488232033658	0.948552316943069	-3.68311464542496
ENSG00000119508	NR4A3	ENSG00000119508	8013	0.961368567719826	0.393124888186165	2.67931987337994	0.0100615884324334	0.99198115045218	-3.36754136162677

01. 差异分析 Summary: DLD1

说明：该表概括当前 LRC vs BULK 比较的样本数、显著上调/下调基因数和阈值设置，用于判断每个细胞系或组合模型中休眠样 LRC 相对 BULK 的总体转录变化强度。

相对路径： `results/ngs/GSE114012/tables/DLD1/DEG/summary.csv`

表格预览：展示全部列与前 21 行。

Dataset	Data_Type	Analysis_Name	Contrast	Samples_Used	Up	Down	Total_Significant_Genes	P_Value_Column	P_Value_Cutoff	LogFC_Cutoff	Microarray_Log2_Transformed
GSE114012	ngs	DLD1	DLD1_vs_BULK	8	541	213	754	P.Value	0.05	0.5	-

01. DLD1: all_genes

说明：该表保留完整差异分析排序结果，可用于追溯每个基因在 LRC/BULK 模型中的 logFC、显著性和排序位置，是后续候选 TF、通路和药物重定位输入的基础。

相对路径： results/ngs/GSE114012/tables/DLD1/DEG/all_genes.csv

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000288825	H2AC18	ENSG00000288825	8337	1.6737983753546	3.20662179580149	15.0539868276589	1.35159797435202e-08	9.49241781488586e-05	9.06866601005878
ENSG00000288859	H2AC19	ENSG00000288859	723790	1.68070082384905	3.19985541490235	14.9360833664103	1.46623691919769e-08	9.49241781488586e-05	9.00687525868469
ENSG00000123384	LRP1	ENSG00000123384	4035	0.669146935195844	6.63333681931637	14.1148065251509	2.62904619079931e-08	0.000113469633594898	9.74505095949846
ENSG00000160179	ABCG1	ENSG00000160179	9619	1.75263706690956	3.64968532444762	13.0816460461447	5.73435252410657e-08	0.000168104376284118	8.35328954100217
ENSG00000136450	SRSF1	ENSG00000136450	6426	-0.589983372070909	7.72047828396787	-12.8403111301534	6.93484874839382e-08	0.000168104376284118	8.79536007745473
ENSG00000163817	SLC6A20	ENSG00000163817	54716	0.649844078335462	6.02983879934906	12.6705048102641	7.94239058458554e-08	0.000168104376284118	8.6577326359965
ENSG00000228691	NEU1	ENSG00000228691	4758	0.78625404417266	5.60811210556021	12.5037944676976	9.08812661406258e-08	0.000168104376284118	8.51317724786376
ENSG00000143815	LBR	ENSG00000143815	3930	-0.521319188027129	6.60768436249975	-11.9859774878361	1.39550631501017e-07	0.000225833299317671	8.10162683568546
ENSG00000166311	SMPD1	ENSG00000166311	6609	0.700667362822073	5.5113637406664	11.84724282686	1.56974026402459e-07	0.000225833299317671	7.98121116215307
ENSG00000277753	MUC20	ENSG00000277753	200958	0.53541326454965	6.63859130699991	11.4437810766322	2.22549949316351e-07	0.000288157674374811	7.63399913609488
ENSG00000130513	GDF15	ENSG00000130513	9518	0.722678717266076	6.62177991076453	11.2482253402481	2.64588238814429e-07	0.000311444410560839	7.46100684926429
ENSG00000276358	PLEKHM1	ENSG00000276358	9842	0.478633980992921	7.63737839754914	11.0380534151847	3.19584852983267e-07	0.000344832056368945	7.26321161618715
ENSG00000168003	SLC3A2	ENSG00000168003	6520	0.634594556602265	8.57470474823492	10.83115890642	3.8603838621999e-07	0.000378265047384138	7.08662254606513
ENSG00000051108	HERPUD1	ENSG00000051108	9709	0.5483753879604	6.7526859225706	10.7604032112199	4.12091114386612e-07	0.000378265047384138	7.0112032365038
ENSG00000163435	ELF3	ENSG00000163435	1999	0.733313764899572	6.38091861236348	10.6941731213768	4.38212520139178e-07	0.000378265047384138	6.95774959796001
ENSG00000077312	SNRPA	ENSG00000077312	6626	-0.653941700849514	5.7720868647973	-10.5594104054726	4.97081760584439e-07	0.000391842442788943	6.84756287637632
ENSG00000128050	PAICS	ENSG00000128050	10606	-0.502613078702901	8.05305306319258	-10.5229052820504	5.14467217130988e-07	0.000391842442788943	6.78599821097825
ENSG00000144354	CDCA7	ENSG00000144354	83879	-0.811342975389484	4.68802476763086	-10.3672633297833	5.96351088263571e-07	0.000410694510473681	6.65230247215091
ENSG00000074527	NTN4	ENSG00000074527	59277	0.752801792276687	5.07970591357115	10.3562535003337	6.02656448795176e-07	0.000410694510473681	6.66110893013925
ENSG00000090615	GOLGA3	ENSG00000090615	2802	0.384930254141882	7.00026582284011	10.164233364533	7.25089130209726e-07	0.000452553994920043	6.43301424347392
ENSG00000165376	CLDN2	ENSG00000165376	9075	-1.40818237904057	4.6838561587445	-10.1256712740365	7.5278870889364e-07	0.000452553994920043	6.42256676585357

01. DLD1: significant_genes

说明：该表按统一阈值筛选显著差异基因，代表当前模型中最直接的休眠样 LRC 转录特征，并作为交集分析和火山图展示的核心输入。

相对路径： results/ngs/GSE114012/tables/DLD1/DEG/significant_genes.csv

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000288825	H2AC18	ENSG00000288825	8337	1.6737983753546	3.20662179580149	15.0539868276589	1.35159797435202e-08	9.49241781488586e-05	9.06866601005878
ENSG00000288859	H2AC19	ENSG00000288859	723790	1.68070082384905	3.19985541490235	14.9360833664103	1.46623691919769e-08	9.49241781488586e-05	9.00687525868469
ENSG00000123384	LRP1	ENSG00000123384	4035	0.669146935195844	6.63333681931637	14.1148065251509	2.62904619079931e-08	0.000113469633594898	9.74505095949846
ENSG00000160179	ABCG1	ENSG00000160179	9619	1.75263706690956	3.64968532444762	13.0816460461447	5.73435252410657e-08	0.000168104376284118	8.35328954100217
ENSG00000136450	SRSF1	ENSG00000136450	6426	-0.589983372070909	7.72047828396787	-12.8403111301534	6.93484874839382e-08	0.000168104376284118	8.79536007745473
ENSG00000163817	SLC6A20	ENSG00000163817	54716	0.649844078335462	6.02983879934906	12.6705048102641	7.94239058458554e-08	0.000168104376284118	8.6577326359965
ENSG00000228691	NEU1	ENSG00000228691	4758	0.78625404417266	5.60811210556021	12.5037944676976	9.08812661406258e-08	0.000168104376284118	8.51317724786376
ENSG00000143815	LBR	ENSG00000143815	3930	-0.521319188027129	6.60768436249975	-11.9859774878361	1.39550631501017e-07	0.000225833299317671	8.10162683568546
ENSG00000166311	SMPD1	ENSG00000166311	6609	0.700667362822073	5.5113637406664	11.84724282686	1.56974026402459e-07	0.000225833299317671	7.98121116215307
ENSG00000277753	MUC20	ENSG00000277753	200958	0.53541326454965	6.63859130699991	11.4437810766322	2.22549949316351e-07	0.000288157674374811	7.63399913609488
ENSG00000130513	GDF15	ENSG00000130513	9518	0.722678717266076	6.62177991076453	11.2482253402481	2.64588238814429e-07	0.000311444410560839	7.46100684926429
ENSG00000168003	SLC3A2	ENSG00000168003	6520	0.634594556602265	8.57470474823492	10.83115890642	3.8603838621999e-07	0.000378265047384138	7.08662254606513
ENSG00000051108	HERPUD1	ENSG00000051108	9709	0.5483753879604	6.7526859225706	10.7604032112199	4.12091114386612e-07	0.000378265047384138	7.0112032365038
ENSG00000163435	ELF3	ENSG00000163435	1999	0.733313764899572	6.38091861236348	10.6941731213768	4.38212520139178e-07	0.000378265047384138	6.95774959796001
ENSG00000077312	SNRPA	ENSG00000077312	6626	-0.653941700849514	5.7720868647973	-10.5594104054726	4.97081760584439e-07	0.000391842442788943	6.84756287637632
ENSG00000128050	PAICS	ENSG00000128050	10606	-0.502613078702901	8.05305306319258	-10.5229052820504	5.14467217130988e-07	0.000391842442788943	6.78599821097825
ENSG00000144354	CDCA7	ENSG00000144354	83879	-0.811342975389484	4.68802476763086	-10.3672633297833	5.96351088263571e-07	0.000410694510473681	6.65230247215091
ENSG00000074527	NTN4	ENSG00000074527	59277	0.752801792276687	5.07970591357115	10.3562535003337	6.02656448795176e-07	0.000410694510473681	6.66110893013925
ENSG00000165376	CLDN2	ENSG00000165376	9075	-1.40818237904057	4.6838561587445	-10.1256712740365	7.5278870889364e-07	0.000452553994920043	6.42256676585357
ENSG00000196878	LAMB3	ENSG00000196878	3914	0.703474021876742	8.73125570896915	10.0648790100083	7.98812795704189e-07	0.000452553994920043	6.35635874445822
ENSG00000053747	LAMA3	ENSG00000053747	3909	0.676924551268898	7.15758101486727	10.0584086875263	8.03888004569122e-07	0.000452553994920043	6.32672392386575

01. 差异分析 Summary: DLD1/HCT15

说明：该表概括当前 LRC vs BULK 比较的样本数、显著上调/下调基因数和阈值设置，用于判断每个细胞系或组合模型中休眠样 LRC 相对 BULK 的总体转录变化强度。

相对路径：`results/ngs/GSE114012/tables/DLD1_HCT15/DEG/summary.csv`

表格预览：展示全部列与前 21 行。

Dataset	Data_Type	Analysis_Name	Contrast	Samples_Used	Up	Down	Total_Significant_Genes	P_Value_Column	P_Value_Cutoff	LogFC_Cutoff	Microarray_Log2_Transformed
GSE114012	ngs	DLD1_HCT15	DLD1+HCT15_vs_BULK	16	57	7	64	P.Value	0.05	0.5	-

01. DLD1/HCT15: all_genes

说明：该表保留完整差异分析排序结果，可用于追溯每个基因在 LRC/BULK 模型中的 logFC、显著性和排序位置，是后续候选 TF、通路和药物重定位输入的基础。

相对路径：`results/ngs/GSE114012/tables/DLD1_HCT15/DEG/all_genes.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000116771	AGMAT	ENSG00000116771	79814	-0.496400781447915	4.5866038979935	-4.19391908607639	0.000695344677368495	0.999937645359275	-3.14288102350188
ENSG00000177192	PUS1	ENSG00000177192	80324	-0.367731292519643	5.14318676635506	-4.1051753677003	0.000836878637218709	0.999937645359275	-3.0507555122286
ENSG00000115107	STEAP3	ENSG00000115107	55240	-0.357557095627065	5.59490522656537	-4.0854365091174	0.00087214289351462	0.999937645359275	-2.98330746703723
ENSG00000122884	P4HA1	ENSG00000122884	5033	0.61751246467013	6.47881751941917	4.03757025656271	0.000964040307842658	0.999937645359275	-2.92101581121668
ENSG00000006652	IFRD1	ENSG00000006652	3475	0.615133313863105	5.74956076614399	3.92167485187032	0.00122926230183359	0.999937645359275	-3.05044577191131
ENSG00000112773	TENT5A	ENSG00000112773	55603	0.608141555538067	3.9530886474613	3.89378739214871	0.00130340922819239	0.999937645359275	-3.46353228338398
ENSG00000011422	PLAUR	ENSG00000011422	5329	0.50748099126209	5.26048247646179	3.87814656251723	0.00134695325532521	0.999937645359275	-3.14288923291662
ENSG00000156804	FBXO32	ENSG00000156804	114907	0.553626440522994	4.8466116740381	3.85420111247964	0.00141647648656496	0.999937645359275	-3.23521979612025
ENSG00000100292	HMOX1	ENSG00000100292	3162	0.660413830713756	3.10447501859139	3.82526106236372	0.00150534598473297	0.999937645359275	-3.75496669323849
ENSG00000111729	CLEC4A	ENSG00000111729	50856	1.04566012742366	-0.782125362368796	3.79974993288721	0.00158833234183762	0.999937645359275	-4.5011189804313
ENSG00000168003	SLC3A2	ENSG00000168003	6520	0.640919351250224	7.15332742190588	3.78076033222623	0.00165308037946813	0.999937645359275	-3.04395760809539
ENSG00000114019	AMOTL2	ENSG00000114019	51421	0.626052683450643	5.36771444691783	3.75661330375836	0.00173926239332088	0.999937645359275	-3.18911895377546
ENSG00000070785	EIF2B3	ENSG00000070785	8891	-0.245942674026067	4.0285183913504	-3.75183013559351	0.00175686250511786	0.999937645359275	-3.48279890223469
ENSG00000164266	SPINK1	ENSG00000164266	6690	1.64822382055825	3.32614148831384	3.74015472461351	0.00180057861904124	0.999937645359275	-3.76476812758316
ENSG00000189308	LIN54	ENSG00000189308	132660	0.293658749654689	4.39636189565989	3.69594069409044	0.00197626612013572	0.999937645359275	-3.40673940026404
ENSG00000204421	LY6G6C	ENSG00000204421	80740	0.66996127807905	2.15924619765812	3.63858936119333	0.00223003332511677	0.999937645359275	-4.05198632464606
ENSG00000281766	RYBP	ENSG00000281766	23429	0.294321437460417	5.5200480951792	3.48268050262425	0.00309717658958257	0.999937645359275	-3.308684082424
ENSG00000137193	PIM1	ENSG00000137193	5292	0.294484807559503	6.04394262932473	3.47897564823768	0.00312144067574187	0.999937645359275	-3.2565968829049
ENSG00000160401	CFAP157	ENSG00000160401	286207	-0.339008287560789	3.25436044044499	-3.46353859783731	0.00322459843421125	0.999937645359275	-3.80939246630071
ENSG00000142871	CCN1	ENSG00000142871	3491	1.51258676840781	0.4536957442861	3.4607829831171	0.00324336692618537	0.999937645359275	-4.42688475684026
ENSG00000132823	OSER1	ENSG00000132823	51526	0.561789345782825	5.89744047681087	3.45708270065737	0.00326874085964825	0.999937645359275	-3.28528747357401

01. DLD1/HCT15: significant_genes

说明：该表按统一阈值筛选显著差异基因，代表当前模型中最直接的休眠样 LRC 转录特征，并作为交集分析和火山图展示的核心输入。

相对路径： results/ngs/GSE114012/tables/DLD1_HCT15/DEG/significant_genes.csv

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG000000122884	P4HA1	ENSG000000122884	5033	0.61751246467013	6.47881751941917	4.03757025656271	0.000964040307842658	0.999937645359275	-2.92101581121668
ENSG000000006652	IFRD1	ENSG000000006652	3475	0.615133313863105	5.74956076614399	3.92167485187032	0.00122926230183359	0.999937645359275	-3.05044577191131
ENSG000000112773	TENT5A	ENSG000000112773	55603	0.608141555538067	3.9530886474613	3.89378739214871	0.00130340922819239	0.999937645359275	-3.46353228338398
ENSG00000011422	PLAUR	ENSG00000011422	5329	0.50748099126209	5.26048247646179	3.87814656251723	0.00134695325532521	0.999937645359275	-3.14288923291662
ENSG000000156804	FBXO32	ENSG000000156804	114907	0.5536264440522994	4.8466116740381	3.85420111247964	0.00141647648656496	0.999937645359275	-3.23521979612025
ENSG000000100292	HMOX1	ENSG000000100292	3162	0.660413830713756	3.10447501859139	3.82526106236372	0.00150534598473297	0.999937645359275	-3.75496669323849
ENSG000000111729	CLEC4A	ENSG000000111729	50856	1.04566012742366	-0.782125362368796	3.79974993288721	0.00158833234183762	0.999937645359275	-4.5011189804313
ENSG000000168003	SLC3A2	ENSG000000168003	6520	0.640919351250224	7.15332742190588	3.78076033222623	0.00165308037946813	0.999937645359275	-3.04395760809539
ENSG000000114019	AMOTL2	ENSG000000114019	51421	0.626052683450643	5.36771444691783	3.75661330375836	0.00173926239332088	0.999937645359275	-3.18911895377546
ENSG000000164266	SPINK1	ENSG000000164266	6690	1.64822382055825	3.32614148831384	3.74015472461351	0.00180057861904124	0.999937645359275	-3.76476812758316
ENSG000000204421	LY6G6C	ENSG000000204421	80740	0.66996127807905	2.15924619765812	3.63858936119333	0.00223003332511677	0.999937645359275	-4.05198632464606
ENSG000000142871	CCN1	ENSG000000142871	3491	1.51258676840781	0.4536957442861	3.4607829831171	0.00324336692618537	0.999937645359275	-4.42688475684026
ENSG000000132823	OSER1	ENSG000000132823	51526	0.561789345782825	5.89744047681087	3.45708270065737	0.00326874085964825	0.999937645359275	-3.28528747357401
ENSG000000118515	SGK1	ENSG000000118515	6446	0.537286085606935	3.1683783701682	3.4299995258143	0.00346057653109383	0.999937645359275	-3.85009509085033
ENSG000000144655	CSRNP1	ENSG000000144655	64651	0.661236839780738	4.2655041941979	3.36756335986976	0.00394657024698275	0.999937645359275	-3.59533567614989
ENSG000000184205	TSPYL2	ENSG000000184205	64061	0.987565148787777	3.89475273973002	3.32785486197777	0.00429033049310044	0.999937645359275	-3.71335686062883
ENSG000000198515	CNGA1	ENSG000000198515	1259	0.644816763937567	1.24964289244471	3.32106840295072	0.00435198427224666	0.999937645359275	-4.29737569225973
ENSG000000124762	CDKN1A	ENSG000000124762	1026	0.507473996699959	5.22324456337022	3.30782453888504	0.00447484316775267	0.999937645359275	-3.44357101370566
ENSG000000156398	SFXN2	ENSG000000156398	118980	-0.511856468522996	1.77835254448683	-3.26209430213196	0.00492607813790303	0.999937645359275	-4.19583351785795
ENSG000000160180	TFF3	ENSG000000160180	7033	0.566250379995876	7.79034165758358	3.25504437948031	0.00499953814079857	0.999937645359275	-3.36093875889493
ENSG000000175197	DDIT3	ENSG000000175197	1649	1.27701743630825	4.44438965377686	3.23522519565412	0.00521190716428547	0.999937645359275	-3.64389544919772

01. 差异分析 Summary: DLD1/HCT15/SW48

说明：该表概括当前 LRC vs BULK 比较的样本数、显著上调/下调基因数和阈值设置，用于判断每个细胞系或组合模型中休眠样 LRC 相对 BULK 的总体转录变化强度。

相对路径： `results/ngs/GSE114012/tables/DLD1_HCT15_SW48/DEG/summary.csv`

表格预览：展示全部列与前 21 行。

Dataset	Data_Type	Analysis_Name	Contrast	Samples_Used	Up	Down	Total_Significant_Genes	P_Value_Column	P_Value_Cutoff	LogFC_Cutoff	Microarray_Log2_Transformed
GSE114012	ngs	DLD1_HCT15_SW48	DLD1+HCT15+SW48_vs_BULK	24	94	8	102	P.Value	0.05	0.5	-

01. DLD1/HCT15/SW48: all_genes

说明：该表保留完整差异分析排序结果，可用于追溯每个基因在 LRC/BULK 模型中的 logFC、显著性和排序位置，是后续候选 TF、通路和药物重定位输入的基础。

相对路径： results/ngs/GSE114012/tables/DLD1_HCT15_SW48/DEG/all_genes.csv

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000087589	CASS4	ENSG00000087589	57091	1.21120386936873	0.359755005268495	7.18358492081321	1.96815324362144e-07	0.00253694953102804	2.2409038941639
ENSG00000124107	SLPI	ENSG00000124107	6590	1.27469109083693	3.17259123018746	5.67676827463132	7.48951888250302e-06	0.0482699491977319	3.29687318426716
ENSG00000159082	SYNJ1	ENSG00000159082	8867	0.510163883974951	4.09808709246395	5.42809875908269	1.39609181683536e-05	0.0599854117300258	3.09070798957141
ENSG00000292358	GTPBP6	ENSG00000292358	8225	-0.291543662353561	5.87206297022075	-5.2451920398631	2.21362501747989e-05	0.0606594104232789	2.78674180200486
ENSG00000086548	CEACAM6	ENSG00000086548	4680	1.74347789815018	0.63186757044775	5.12055430097328	3.03410613556382e-05	0.0606594104232789	0.264545212961069
ENSG00000120885	CLU	ENSG00000120885	1191	0.854490139003905	7.53355270153405	5.05129208652447	3.61639134679952e-05	0.0606594104232789	2.33278441940098
ENSG00000280789	PAGR1	ENSG00000280789	79447	-0.265740876159433	5.7901062475462	-5.02840080774023	3.83262205941147e-05	0.0606594104232789	2.27992105185593
ENSG00000125775	SDCBP2	ENSG00000125775	27111	0.875088203826344	1.77587005594115	4.96782447639329	4.46975350450178e-05	0.0606594104232789	1.19851232274781
ENSG00000144579	CTDSP1	ENSG00000144579	58190	-0.304650463945078	6.96950747868497	-4.94969479934092	4.68041858428013e-05	0.0606594104232789	2.09626263222225
ENSG00000147274	RBMX	ENSG00000147274	27316	-0.25286457778393	7.61977866524378	-4.94755542392094	4.70592788388509e-05	0.0606594104232789	2.0901990321893
ENSG00000103495	MAZ	ENSG00000103495	4150	-0.288673317754424	7.270465502827	-4.84848162510587	6.0539422588787e-05	0.0709411961063149	1.85824768956248
ENSG00000118503	TNFAIP3	ENSG00000118503	7128	0.701047447962078	2.70673388639959	4.76860544785874	7.41866032580931e-05	0.0752435825778732	1.31375681260624
ENSG00000104325	DECR1	ENSG00000104325	1666	-0.359634637320958	5.4818399269116	-4.7597116382693	7.58856922817961e-05	0.0752435825778732	1.64979290383751
ENSG00000283741	SMCR8	ENSG00000283741	140775	0.219296442377881	6.24412831210502	4.67693004027955	9.36972303092474e-05	0.0862683784775856	1.45288953726875
ENSG00000155744	HYCC2	ENSG00000155744	285172	0.405676239023841	4.5995617636913	4.55064884544191	0.000129262955443717	0.105104114165313	1.1562413499552
ENSG00000090615	GOLGA3	ENSG00000090615	2802	0.242871519267588	7.02368373608467	4.51530046100851	0.000141447569859786	0.105104114165313	1.07225895178003
ENSG00000138764	CCNG2	ENSG00000138764	901	0.370839517586568	6.0795485591222	4.51085213546323	0.00014306009487915	0.105104114165313	1.06202661707903
ENSG00000124201	ZNFX1	ENSG00000124201	57169	0.326419540543414	6.39061910683048	4.5008035056374	0.000146770679206799	0.105104114165313	1.03643318901372
ENSG00000132510	KDM6B	ENSG00000132510	23135	0.253918678042418	5.61505610867772	4.46400741815524	0.000161197744270814	0.109359943350041	0.956288953174293
ENSG00000164687	FABP5	ENSG00000164687	2171	-0.389872741436999	4.10335972023549	-4.44145498659001	0.000170731430294807	0.110036406825003	0.889236855381561
ENSG00000162772	ATF3	ENSG00000162772	467	1.47929162089823	3.30987044307823	4.416204079105	0.00018207513501573	0.11175945192156	0.720762300669653

01. DLD1/HCT15/SW48: significant_genes

说明：该表按统一阈值筛选显著差异基因，代表当前模型中最直接的休眠样 LRC 转录特征，并作为交集分析和火山图展示的核心输入。

相对路径： results/ngs/GSE114012/tables/DLD1_HCT15_SW48/DEG/significant_genes.csv

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000087589	CASS4	ENSG00000087589	57091	1.21120386936873	0.359755005268495	7.18358492081321	1.96815324362144e-07	0.00253694953102804	2.2409038941639
ENSG00000124107	SLPI	ENSG00000124107	6590	1.27469109083693	3.17259123018746	5.67676827463132	7.48951888250302e-06	0.0482699491977319	3.29687318426716
ENSG00000159082	SYNJ1	ENSG00000159082	8867	0.510163883974951	4.09808709246395	5.42809875908269	1.39609181683536e-05	0.0599854117300258	3.09070798957141
ENSG00000086548	CEACAM6	ENSG00000086548	4680	1.74347789815018	0.63186757044775	5.12055430097328	3.03410613556382e-05	0.0606594104232789	0.264545212961069
ENSG00000120885	CLU	ENSG00000120885	1191	0.854490139003905	7.53355270153405	5.05129208652447	3.61639134679952e-05	0.0606594104232789	2.33278441940098
ENSG00000125775	SDCBP2	ENSG00000125775	27111	0.875088203826344	1.77587005594115	4.96782447639329	4.46975350450178e-05	0.0606594104232789	1.19851232274781
ENSG00000118503	TNFAIP3	ENSG00000118503	7128	0.701047447962078	2.70673388639959	4.76860544785874	7.41866032580931e-05	0.0752435825778732	1.31375681260624
ENSG00000162772	ATF3	ENSG00000162772	467	1.47929162089823	3.30987044307823	4.416204079105	0.00018207513501573	0.11175945192156	0.720762300669653
ENSG00000131480	AOC2	ENSG00000131480	314	0.935444929065818	1.31867829597724	4.35156803723763	0.000214655360889326	0.114540689070376	-0.1611330775038
ENSG00000196611	MMP1	ENSG00000196611	4312	2.73884043732475	-0.967344951008732	4.03267784651107	0.000482576765098057	0.168119310867945	-2.08806223220993
ENSG00000162733	DDR2	ENSG00000162733	4921	0.693999328784209	0.389294058795494	3.9205822568603	0.000640711352303452	0.192064403050965	-1.40318799142194
ENSG00000138166	DUSP5	ENSG00000138166	1847	1.01979765549424	5.51521441366224	3.88619043917711	0.000698787919699304	0.200456971239222	-0.387360945221626
ENSG00000197409	H3C4	ENSG00000197409	8351	0.890321194123998	0.783427255144521	3.76873800989462	0.00093905177175683	0.227333116286445	-1.39339024843509
ENSG00000131471	AOC3	ENSG00000131471	8639	0.95339605693719	0.4137290058453	3.70130213480386	0.00111200012202739	0.238894692882218	-1.69672363924861
ENSG00000173110	HSPA6	ENSG00000173110	3310	1.32498423784302	0.349088977710574	3.6448488326179	0.00128053178508057	0.249498257426766	-1.83023651282581
ENSG00000161681	SHANK1	ENSG00000161681	50944	0.656577190496357	1.89110462462165	3.63081916996321	0.00132615648940095	0.249498257426766	-1.14705923483305
ENSG00000140961	OSGIN1	ENSG00000140961	29948	0.500958949761153	3.2551477949111	3.58602611979069	0.00148274040740662	0.261815395225635	-1.02719275830988
ENSG00000130513	GDF15	ENSG00000130513	9518	1.0851554340722	6.36167420442492	3.53100805351904	0.00169996212030286	0.280929637573126	-1.21922997746803
ENSG00000107738	VSIR	ENSG00000107738	64115	0.835027179871227	2.36873677337309	3.51159358164574	0.00178380102930439	0.285688652380635	-1.27250824346522
ENSG00000163817	SLC6A20	ENSG00000163817	54716	0.920146126457797	5.97765576881827	3.49703825904174	0.00184929016342962	0.285688652380635	-1.29159797874327
ENSG00000143178	TBX19	ENSG00000143178	9095	0.525321872528624	1.58120264078008	3.4863828544066	0.00189871353839516	0.287934323646043	-1.4908144850107

01. 差异分析 Summary: HCT15

说明：该表概括当前 LRC vs BULK 比较的样本数、显著上调/下调基因数和阈值设置，用于判断每个细胞系或组合模型中休眠样 LRC 相对 BULK 的总体转录变化强度。

相对路径： `results/ngs/GSE114012/tables/HCT15/DEG/summary.csv`

表格预览：展示全部列与前 21 行。

Dataset	Data_Type	Analysis_Name	Contrast	Samples_Used	Up	Down	Total_Significant_Genes	P_Value_Column	P_Value_Cutoff	LogFC_Cutoff	Microarray_Log2_Transformed
GSE114012	ngs	HCT15	HCT15_vs_BULK	8	362	161	523	P.Value	0.05	0.5	-

01. HCT15: all_genes

说明：该表保留完整差异分析排序结果，可用于追溯每个基因在 LRC/BULK 模型中的 logFC、显著性和排序位置，是后续候选 TF、通路和药物重定位输入的基础。

相对路径：`results/ngs/GSE114012/tables/HCT15/DEG/all_genes.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG000000137440	FGFBP1	ENSG000000137440	9982	-0.729382561734184	5.9210978631537	-9.90869498319374	1.31066295226287e-06	0.0108985594644346	5.90055358640995
ENSG000000182578	CSF1R	ENSG000000182578	1436	2.57498680890415	-0.47368902605195	9.66893384328526	1.65117179977799e-06	0.0108985594644346	2.41664373912222
ENSG000000134321	RSAD2	ENSG000000134321	91543	1.95775079674493	1.58781823197426	8.51964220631571	5.35117472467346e-06	0.0230353946901153	3.36433420039665
ENSG000000137501	SYTL2	ENSG000000137501	54843	-0.814445037938041	6.51953066705869	-7.94692327543605	1.00751635017657e-05	0.0230353946901153	3.93233369550282
ENSG000000124107	SLPI	ENSG000000124107	6590	1.80181890412888	3.36532875338617	7.91840728066626	1.04068541066609e-05	0.0230353946901153	3.7050457244412
ENSG000000213246	SUPT4H1	ENSG000000213246	6827	-0.72748441049869	9.19844120607297	-7.76390929412066	1.24220659695992e-05	0.0230353946901153	3.7328394960341
ENSG000000083799	CYLD	ENSG000000083799	1540	0.62405320302401	5.31261813932099	7.59867024199813	1.50543980077551e-05	0.0230353946901153	3.54795814470888
ENSG000000110721	CHKA	ENSG000000110721	1119	0.585897648270165	5.29690059539288	7.58323859028009	1.53294181369006e-05	0.0230353946901153	3.53039225806102
ENSG000000013588	GPRC5A	ENSG000000013588	9052	0.869349357156766	7.91978976344755	7.56265397695755	1.57047611704445e-05	0.0230353946901153	3.49433144212656
ENSG000000104325	DECRC1	ENSG000000104325	1666	-0.603422809894652	5.56725201794052	-7.24320756789884	2.30010768313224e-05	0.0257378548912952	3.12871965813409
ENSG000000225553	PHF1	ENSG000000225553	5252	0.634524255491741	5.92827944678821	7.23063874475396	2.33546395798327e-05	0.0257378548912952	3.10645902594441
ENSG000000127863	TNFRSF19	ENSG000000127863	55504	-0.683114132030092	5.64551505155675	-7.20993430782328	2.39499170362716e-05	0.0257378548912952	3.08739307565794
ENSG000000132823	OSER1	ENSG000000132823	51526	0.641543665971061	5.82938896584078	7.1634808426753	2.53459672439086e-05	0.0257378548912952	3.02742050925747
ENSG000000116132	PRRX1	ENSG000000116132	5396	-1.06585585041068	2.33067317208611	-6.96899869975854	3.22206156717145e-05	0.0303817391058788	2.52429676550831
ENSG000000196611	MMP1	ENSG000000196611	4312	3.03434016564328	-1.16319021218707	6.8621964474204	3.68313043263673e-05	0.0324140032274917	0.942782730330908
ENSG000000165272	AQP3	ENSG000000165272	360	0.819381254500359	5.39716792732305	6.80654476997781	3.95115749489799e-05	0.0325995188063428	2.59993764028578
ENSG000000107738	VSIR	ENSG000000107738	64115	1.13802569042159	2.3662688518295	6.74296375525147	4.28334367584577e-05	0.0332614234499059	2.2961955914556
ENSG000000169155	ZBTB43	ENSG000000169155	23099	0.781234392792199	4.84337592707157	6.59019427357573	5.21101591929577e-05	0.0347154177656026	2.34344513627565
ENSG000000196352	CD55	ENSG000000196352	1604	0.406319949484558	6.98286870241085	6.56991190446263	5.34961672944816e-05	0.0347154177656026	2.26317716251193
ENSG000000162772	ATF3	ENSG000000162772	467	1.24338585742293	4.2450824266744	6.51804111509626	5.72245266635181e-05	0.0347154177656026	2.26133374160009
ENSG000000078081	LAMP3	ENSG000000078081	27074	1.13324013832028	3.73184673913899	6.4662389510687	6.12284391223567e-05	0.0347154177656026	2.18594254887515

01. HCT15: significant_genes

说明：该表按统一阈值筛选显著差异基因，代表当前模型中最直接的休眠样 LRC 转录特征，并作为交集分析和火山图展示的核心输入。

相对路径： results/ngs/GSE114012/tables/HCT15/DEG/significant_genes.csv

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000137440	FGFBP1	ENSG00000137440	9982	-0.729382561734184	5.9210978631537	-9.90869498319374	1.31066295226287e-06	0.0108985594644346	5.90055358640995
ENSG00000182578	CSF1R	ENSG00000182578	1436	2.57498680890415	-0.47368902605195	9.66893384328526	1.65117179977799e-06	0.0108985594644346	2.41664373912222
ENSG00000134321	RSAD2	ENSG00000134321	91543	1.95775079674493	1.58781823197426	8.51964220631571	5.35117472467346e-06	0.0230353946901153	3.36433420039665
ENSG00000137501	SYTL2	ENSG00000137501	54843	-0.814445037938041	6.51953066705869	-7.94692327543605	1.00751635017657e-05	0.0230353946901153	3.93233369550282
ENSG00000124107	SLPI	ENSG00000124107	6590	1.80181890412888	3.36532875338617	7.91840728066626	1.04068541066609e-05	0.0230353946901153	3.7050457244412
ENSG00000213246	SUPT4H1	ENSG00000213246	6827	-0.72748441049869	9.19844120607297	-7.76390929412066	1.24220659695992e-05	0.0230353946901153	3.7328394960341
ENSG00000083799	CYLD	ENSG00000083799	1540	0.62405320302401	5.31261813932099	7.59867024199813	1.50543980077551e-05	0.0230353946901153	3.54795814470888
ENSG00000110721	CHKA	ENSG00000110721	1119	0.585897648270165	5.29690059539288	7.58323859028009	1.53294181369006e-05	0.0230353946901153	3.53039225806102
ENSG00000013588	GPRC5A	ENSG00000013588	9052	0.869349357156766	7.76178976344755	7.56265397695755	1.57047611704445e-05	0.0230353946901153	3.49433144212656
ENSG00000104325	DECRC1	ENSG00000104325	1666	-0.603422809894652	5.56725201794052	-7.24320756789884	2.30010768313224e-05	0.0257378548912952	3.12871965813409
ENSG00000225553	PHF1	ENSG00000225553	5252	0.634524255491741	5.92827944678821	7.23063874475396	2.33546395798327e-05	0.0257378548912952	3.10645902594441
ENSG00000127863	TNFRSF19	ENSG00000127863	55504	-0.683114132030092	5.64551505155675	-7.20993430782328	2.39499170362716e-05	0.0257378548912952	3.08739307565794
ENSG00000132823	OSER1	ENSG00000132823	51526	0.641543665971061	5.82938896584078	7.1634808426753	2.53459672439086e-05	0.0257378548912952	3.02742050925747
ENSG00000116132	PRRX1	ENSG00000116132	5396	-1.06585585041068	2.33067317208611	-6.96899869975854	3.22206156717145e-05	0.0303817391058788	2.52429676550831
ENSG00000196611	MMP1	ENSG00000196611	4312	3.03434016564328	-1.16319021218707	6.8621964474204	3.68313043263673e-05	0.0324140032274917	0.942782730330908
ENSG00000165272	AQP3	ENSG00000165272	360	0.819381254500359	5.39716792732305	6.80654476997781	3.95115749489799e-05	0.0325995188063428	2.59993764028578
ENSG00000107738	VSIR	ENSG00000107738	64115	1.13802569042159	2.3662688518295	6.74296375525147	4.28334367584577e-05	0.0332614234499059	2.2961955914556
ENSG00000169155	ZBTB43	ENSG00000169155	23099	0.781234392792199	4.84337592707157	6.59019427357573	5.21101591929577e-05	0.0347154177656026	2.34344513627565
ENSG00000162772	ATF3	ENSG00000162772	467	1.24338585742293	4.2450824266744	6.51804111509626	5.72245266635181e-05	0.0347154177656026	2.26133374160009
ENSG00000078081	LAMP3	ENSG00000078081	27074	1.13324013832028	3.73184673913899	6.4662389510687	6.12284391223567e-05	0.0347154177656026	2.18594254887515
ENSG00000105974	CAV1	ENSG00000105974	857	0.510388753207357	4.86838080860424	6.38421583466312	6.81974405696628e-05	0.0347154177656026	2.07811026298555

01. 差异分析 Summary: HT55

说明：该表概括当前 LRC vs BULK 比较的样本数、显著上调/下调基因数和阈值设置，用于判断每个细胞系或组合模型中休眠样 LRC 相对 BULK 的总体转录变化强度。

相对路径： `results/ngs/GSE114012/tables/HT55/DEG/summary.csv`

表格预览：展示全部列与前 21 行。

Dataset	Data_Type	Analysis_Name	Contrast	Samples_Used	Up	Down	Total_Significant_Genes	P_Value_Column	P_Value_Cutoff	LogFC_Cutoff	Microarray_Log2_Transformed
GSE114012	ngs	HT55	HT55_vs_BULK	8	307	60	367	P.Value	0.05	0.5	-

01. HT55: all_genes

说明：该表保留完整差异分析排序结果，可用于追溯每个基因在 LRC/BULK 模型中的 logFC、显著性和排序位置，是后续候选 TF、通路和药物重定位输入的基础。

相对路径：`results/ngs/GSE114012/tables/HT55/DEG/all_genes.csv`

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG000000204421	LY6G6C	ENSG000000204421	80740	1.16298559366389	1.98371311854092	5.60405680315523	0.000150779305196176	0.405693176761372	0.122992335431197
ENSG000000187210	GCNT1	ENSG000000187210	2650	0.71828020096137	5.34424529548926	5.45979488263143	0.000187696482631205	0.405693176761372	1.01065726468172
ENSG00000059804	SLC2A3	ENSG00000059804	6515	1.66518145583407	6.80095289108856	5.3500537469641	0.000222160764810581	0.405693176761372	0.884403744862246
ENSG000000198734	F5	ENSG000000198734	2153	-0.991785205661427	5.81285775269313	-5.23798490400841	0.000264360881971845	0.405693176761372	0.728425175872367
ENSG000000285069	SESN2	ENSG000000285069	83667	0.940276054652189	2.49812737530616	5.16197501253512	0.000297759188862012	0.405693176761372	0.00564894212721434
ENSG000000106038	EVX1	ENSG000000106038	2128	0.687952848892348	3.90445011239384	5.12804674487366	0.000314081850789874	0.405693176761372	0.428153167579627
ENSG000000007952	NOX1	ENSG000000007952	27035	1.60723965508579	2.6998589720155	5.11406516258211	0.000321081131286379	0.405693176761372	-0.0028823692304174
ENSG000000182985	CADM1	ENSG000000182985	23705	1.20395508289979	0.817718090652384	5.10850033609997	0.000323912619907804	0.405693176761372	-1.01880931637682
ENSG000000173262	SLC2A14	ENSG000000173262	144195	1.64952846658131	4.20354022941747	4.99077776909785	0.000390425537104756	0.405693176761372	0.264084311709969
ENSG000000119950	MXI1	ENSG000000119950	4601	0.769903938787007	4.9058986030054	4.97376453295182	0.000401171881615839	0.405693176761372	0.329473613449639
ENSG000000140853	NLR5	ENSG000000140853	84166	-0.58598522055533	4.61462501719449	-4.93736731678053	0.000425223859328897	0.405693176761372	0.263686850628111
ENSG000000153233	PTPRR	ENSG000000153233	5801	0.780626885373714	4.48602828209368	4.82931375069941	0.000506015140450154	0.405693176761372	0.102447894678306
ENSG000000198832	SELENOM	ENSG000000198832	140606	1.99704548561285	-1.20618961278681	4.80596390890353	0.000525510771443461	0.405693176761372	-2.13919951409128
ENSG000000181061	HIGD1A	ENSG000000181061	25994	0.424116012341031	6.73135595155754	4.7963665436247	0.000533751885191591	0.405693176761372	0.11132633944487
ENSG00000059728	MXD1	ENSG00000059728	4084	0.473243680100313	7.01511672250785	4.7526756880605	0.000573025074905971	0.405693176761372	0.0465053214019653
ENSG000000175197	DDIT3	ENSG000000175197	1649	0.801262967544231	3.97870901402782	4.74983731812166	0.000575679463335577	0.405693176761372	-0.0579139788854031
ENSG000000136826	KLF4	ENSG000000136826	9314	0.617346837930349	6.98700673238482	4.73582954973396	0.000588970031569451	0.405693176761372	0.0216100780133237
ENSG000000167723	TRPV3	ENSG000000167723	162514	0.583586840896095	3.18267292287105	4.73182099647349	0.000592832461129839	0.405693176761372	-0.228669299525227
ENSG000000156804	FBXO32	ENSG000000156804	114907	0.685333862156594	4.85805117070455	4.71755171996004	0.000606799209514766	0.405693176761372	-0.0318646614573916
ENSG000000281618	IFITM10	ENSG000000281618	402778	0.672671435474153	2.79679638009832	4.66333272407533	0.000663105116584816	0.413218410046681	-0.424521699531848
ENSG000000118523	CCN2	ENSG000000118523	1490	1.03624209601434	3.04919791494939	4.57062375340252	0.000772469336818967	0.413218410046681	-0.477334102299416

01. HT55: significant_genes

说明：该表按统一阈值筛选显著差异基因，代表当前模型中最直接的休眠样 LRC 转录特征，并作为交集分析和火山图展示的核心输入。

相对路径： results/ngs/GSE114012/tables/HT55/DEG/significant_genes.csv

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG000000204421	LY6G6C	ENSG000000204421	80740	1.16298559366389	1.98371311854092	5.60405680315523	0.000150779305196176	0.405693176761372	0.122992335431197
ENSG000000187210	GCNT1	ENSG000000187210	2650	0.71828020096137	5.34424529548926	5.45979488263143	0.000187696482631205	0.405693176761372	1.01065726468172
ENSG00000059804	SLC2A3	ENSG00000059804	6515	1.66518145583407	6.80095289108856	5.3500537469641	0.000222160764810581	0.405693176761372	0.884403744862246
ENSG000000198734	F5	ENSG000000198734	2153	-0.991785205661427	5.81285775269313	-5.23798490400841	0.000264360881971845	0.405693176761372	0.728425175872367
ENSG000000285069	SESN2	ENSG000000285069	83667	0.940276054652189	2.49812737530616	5.16197501253512	0.000297759188862012	0.405693176761372	0.00564894212721434
ENSG000000106038	EVX1	ENSG000000106038	2128	0.687952848892348	3.90445011239384	5.12804674487366	0.000314081850789874	0.405693176761372	0.428153167579627
ENSG000000007952	NOX1	ENSG000000007952	27035	1.60723965508579	2.6998589720155	5.11406516258211	0.000321081131286379	0.405693176761372	-0.0028823692304174
ENSG000000182985	CADM1	ENSG000000182985	23705	1.20395508289979	0.817718090652384	5.10850033609997	0.000323912619907804	0.405693176761372	-1.01880931637682
ENSG000000173262	SLC2A14	ENSG000000173262	144195	1.64952846658131	4.20354022941747	4.99077776909785	0.000390425537104756	0.405693176761372	0.264084311709969
ENSG000000119950	MXI1	ENSG000000119950	4601	0.769903938787007	4.9058986030054	4.97376453295182	0.000401171881615839	0.405693176761372	0.329473613449639
ENSG000000140853	NLR5	ENSG000000140853	84166	-0.58598522055533	4.61462501719449	-4.93736731678053	0.000425223859328897	0.405693176761372	0.263686850628111
ENSG000000153233	PTPRR	ENSG000000153233	5801	0.780626885373714	4.48602828209368	4.82931375069941	0.000506015140450154	0.405693176761372	0.102447894678306
ENSG000000198832	SELENOM	ENSG000000198832	140606	1.99704548561285	-1.20618961278681	4.80596390890353	0.000525510771443461	0.405693176761372	-2.13919951409128
ENSG000000175197	DDIT3	ENSG000000175197	1649	0.801262967544231	3.97870901402782	4.74983731812166	0.000575679463335577	0.405693176761372	-0.0579139788854031
ENSG000000136826	KLF4	ENSG000000136826	9314	0.617346837930349	6.98700673238482	4.73582954973396	0.000588970031569451	0.405693176761372	0.0216100780133237
ENSG000000167723	TRPV3	ENSG000000167723	162514	0.583586840896095	3.18267292287105	4.73182099647349	0.000592832461129839	0.405693176761372	-0.228669299525227
ENSG000000156804	FBXO32	ENSG000000156804	114907	0.685333862156594	4.85805117070455	4.71755171996004	0.000606799209514766	0.405693176761372	-0.0318646614573916
ENSG000000281618	IFITM10	ENSG000000281618	402778	0.672671435474153	2.79679638009832	4.66333272407533	0.000663105116584816	0.413218410046681	-0.424521699531848
ENSG000000118523	CCN2	ENSG000000118523	1490	1.03624209601434	3.04919791494939	4.57062375340252	0.000772469336818967	0.413218410046681	-0.477334102299416
ENSG000000138449	SLC40A1	ENSG000000138449	30061	0.749984539313608	4.34121062995317	4.48562526789658	0.000889424050997988	0.413218410046681	-0.389626772929558
ENSG000000176928	GCNT4	ENSG000000176928	51301	1.2671316819131	0.160347361906448	4.48457267475127	0.000890983620311554	0.413218410046681	-1.83812612193587

01. 差异分析 Summary: RKO

说明：该表概括当前 LRC vs BULK 比较的样本数、显著上调/下调基因数和阈值设置，用于判断每个细胞系或组合模型中休眠样 LRC 相对 BULK 的总体转录变化强度。

相对路径： `results/ngs/GSE114012/tables/RKO/DEG/summary.csv`

表格预览：展示全部列与前 21 行。

Dataset	Data_Type	Analysis_Name	Contrast	Samples_Used	Up	Down	Total_Significant_Genes	P_Value_Column	P_Value_Cutoff	LogFC_Cutoff	Microarray_Log2_Transformed
GSE114012	ngs	RKO	RKO_vs_BULK	8	1252	627	1879	P.Value	0.05	0.5	-

01. RKO: all_genes

说明：该表保留完整差异分析排序结果，可用于追溯每个基因在 LRC/BULK 模型中的 logFC、显著性和排序位置，是后续候选 TF、通路和药物重定位输入的基础。

相对路径：`results/ngs/GSE114012/tables/RKO/DEG/all_genes.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000099250	NRP1	ENSG00000099250	8829	1.86926861354285	6.86880316503046	24.2932112913324	9.14876737551245e-14	1.10489663594064e-09	21.9254784780179
ENSG00000164938	TP53INP1	ENSG00000164938	94241	2.23220512126351	5.65911967654985	22.6772559806882	2.58853029020612e-13	1.56308401574096e-09	20.8725715408707
ENSG00000144821	MYH15	ENSG00000144821	22989	3.19398786119874	4.42792223299635	21.6353493445297	5.25675567141422e-13	2.11619460812232e-09	19.9945023937424
ENSG00000135318	NT5E	ENSG00000135318	4907	1.66608180825344	4.87795955132123	21.0797848333915	7.77166066952637e-13	2.19607400001204e-09	19.7717814413759
ENSG00000147883	CDKN2B	ENSG00000147883	1030	1.84362849581263	6.69263309115466	20.8606031757717	9.09196820407403e-13	2.19607400001204e-09	19.6646636145114
ENSG00000048392	RRM2B	ENSG00000048392	50484	1.57102520370292	7.13639023404807	20.1411732445007	1.53861237918859e-12	2.83343595336018e-09	19.139737733954
ENSG00000128564	VGF	ENSG00000128564	7425	2.36864130677186	6.16554556915756	20.0536160399756	1.6422995506766e-12	2.83343595336018e-09	19.0753378149619
ENSG00000085117	CD82	ENSG00000085117	3732	3.24615188952593	3.47349647895045	17.5929522917447	1.15389506680072e-11	1.72376883876672e-08	16.8916067775407
ENSG00000105974	CAV1	ENSG00000105974	857	1.41915878091584	4.96628547381708	17.4113269250829	1.34552480067727e-11	1.72376883876672e-08	16.9843367170902
ENSG00000214357	NEURL1B	ENSG00000214357	54492	2.53353620257524	2.85101337832356	17.3420214187176	1.42731542499522e-11	1.72376883876672e-08	16.5677966123604
ENSG00000282830	CALB2	ENSG00000282830	794	-1.08897978888935	6.77642622126115	-16.9688014355849	1.96856278732286e-11	2.16130298022711e-08	16.5687791853295
ENSG00000005884	ITGA3	ENSG00000005884	3675	1.42921262206703	7.31635865022663	16.3700885435322	3.34227123006211e-11	3.363717470455e-08	16.02889674885
ENSG00000172667	ZMAT3	ENSG00000172667	64393	1.41390120201257	6.51996574986679	15.9457853667007	4.91563971320645e-11	4.30771657207938e-08	15.644668488248
ENSG00000165891	E2F7	ENSG00000165891	144455	1.24819327905891	6.06295765863978	15.9286832096807	4.99362689485065e-11	4.30771657207938e-08	15.6387983377316
ENSG00000164287	CDC20B	ENSG00000164287	166979	5.51066903588796	1.70764356134445	15.138058105515	1.05145212172742e-10	8.46559151606801e-08	13.5375108745588
ENSG00000114850	SSR3	ENSG00000114850	6747	-1.27451062142177	9.07475430314222	-14.9385305625366	1.27571433101979e-10	9.18899239233051e-08	14.6859615527364
ENSG00000169282	KCNAB1	ENSG00000169282	7881	-1.26103746284906	7.67149004007517	-14.9243511831585	1.29347412991321e-10	9.18899239233051e-08	14.6455360803386
ENSG00000117461	PIK3R3	ENSG00000117461	8503	1.42994609651171	4.43593247791651	14.6030516342901	1.77488791132391e-10	1.15191937809818e-07	14.4173442181485
ENSG00000125398	SOX9	ENSG00000125398	6662	1.94712942636892	4.21333139977524	14.5821147813009	1.81224378437239e-10	1.15191937809818e-07	14.4029276779092
ENSG00000140379	BCL2A1	ENSG00000140379	597	-1.1574869749361	5.66103933513363	-14.4794458274617	2.00788539196159e-10	1.17397618480759e-07	14.2350410513703
ENSG00000184205	TSPYL2	ENSG00000184205	64061	1.48635387200305	5.88792403696203	14.4629463911397	2.04135959931767e-10	1.17397618480759e-07	14.212877472322

01. RKO: significant_genes

说明: 该表按统一阈值筛选显著差异基因, 代表当前模型中最直接的休眠样 LRC 转录特征, 并作为交集分析和火山图展示的核心输入。

相对路径: results/ngs/GSE114012/tables/RKO/DEG/significant_genes.csv

表格预览: 展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000099250	NRP1	ENSG00000099250	8829	1.86926861354285	6.86880316503046	24.2932112913324	9.14876737551245e-14	1.10489663594064e-09	21.9254784780179
ENSG00000164938	TP53INP1	ENSG00000164938	94241	2.23220512126351	5.65911967654985	22.6772559806882	2.58853029020612e-13	1.56308401574096e-09	20.8725715408707
ENSG00000144821	MYH15	ENSG00000144821	22989	3.19398786119874	4.42792223299635	21.6353493445297	5.25675567141422e-13	2.11619460812232e-09	19.9945023937424
ENSG00000135318	NT5E	ENSG00000135318	4907	1.66608180825344	4.87795955132123	21.0797848333915	7.77166066952637e-13	2.19607400001204e-09	19.7717814413759
ENSG00000147883	CDKN2B	ENSG00000147883	1030	1.84362849581263	6.69263309115466	20.8606031757717	9.09196820407403e-13	2.19607400001204e-09	19.6646636145114
ENSG00000048392	RRM2B	ENSG00000048392	50484	1.57102520370292	7.13639023404807	20.1411732445007	1.53861237918859e-12	2.83343595336018e-09	19.139737733954
ENSG00000128564	VGF	ENSG00000128564	7425	2.36864130677186	6.16554556915756	20.0536160399756	1.6422995506766e-12	2.83343595336018e-09	19.0753378149619
ENSG00000085117	CD82	ENSG00000085117	3732	3.24615188952593	3.47349647895045	17.5929522917447	1.15389506680072e-11	1.72376883876672e-08	16.8916067775407
ENSG00000105974	CAV1	ENSG00000105974	857	1.41915878091584	4.96628547381708	17.4113269250829	1.34552480067727e-11	1.72376883876672e-08	16.9843367170902
ENSG00000214357	NEURL1B	ENSG00000214357	54492	2.53353620257524	2.85101337832356	17.3420214187176	1.42731542499522e-11	1.72376883876672e-08	16.5677966123604
ENSG00000282830	CALB2	ENSG00000282830	794	-1.08897978888935	6.77642622126115	-16.9688014355849	1.96856278732286e-11	2.16130298022711e-08	16.5687791853295
ENSG00000005884	ITGA3	ENSG00000005884	3675	1.42921262206703	7.31635865022663	16.3700885435322	3.34227123006211e-11	3.363717470455e-08	16.02889674885
ENSG00000172667	ZMAT3	ENSG00000172667	64393	1.41390120201257	6.51996574986679	15.9457853667007	4.91563971320645e-11	4.30771657207938e-08	15.644668488248
ENSG00000165891	E2F7	ENSG00000165891	144455	1.24819327905891	6.06295765863978	15.9286832096807	4.99362689485065e-11	4.30771657207938e-08	15.6387983377316
ENSG00000164287	CDC20B	ENSG00000164287	166979	5.51066903588796	1.70764356134445	15.138058105515	1.05145212172742e-10	8.46559151606801e-08	13.5375108745588
ENSG00000114850	SSR3	ENSG00000114850	6747	-1.27451062142177	9.07475430314222	-14.9385305625366	1.27571433101979e-10	9.18899239233051e-08	14.6859615527364
ENSG00000169282	KCNAB1	ENSG00000169282	7881	-1.26103746284906	7.67149004007517	-14.9243511831585	1.29347412991321e-10	9.18899239233051e-08	14.6455360803386
ENSG00000117461	PIK3R3	ENSG00000117461	8503	1.42994609651171	4.43593247791651	14.6030516342901	1.77488791132391e-10	1.15191937809818e-07	14.4173442181485
ENSG00000125398	SOX9	ENSG00000125398	6662	1.94712942636892	4.21333139977524	14.5821147813009	1.81224378437239e-10	1.15191937809818e-07	14.4029276779092
ENSG00000140379	BCL2A1	ENSG00000140379	597	-1.1574869749361	5.66103933513363	-14.4794458274617	2.00788539196159e-10	1.17397618480759e-07	14.2350410513703
ENSG00000184205	TSPYL2	ENSG00000184205	64061	1.48635387200305	5.88792403696203	14.4629463911397	2.04135959931767e-10	1.17397618480759e-07	14.212877472322

01. 差异分析 Summary: SW48

说明：该表概括当前 LRC vs BULK 比较的样本数、显著上调/下调基因数和阈值设置，用于判断每个细胞系或组合模型中休眠样 LRC 相对 BULK 的总体转录变化强度。

相对路径： `results/ngs/GSE114012/tables/SW48/DEG/summary.csv`

表格预览：展示全部列与前 21 行。

Dataset	Data_Type	Analysis_Name	Contrast	Samples_Used	Up	Down	Total_Significant_Genes	P_Value_Column	P_Value_Cutoff	LogFC_Cutoff	Microarray_Log2_Transformed
GSE114012	ngs	SW48	SW48_vs_BULK	8	353	96	449	P.Value	0.05	0.5	-

01. SW48: all_genes

说明：该表保留完整差异分析排序结果，可用于追溯每个基因在 LRC/BULK 模型中的 logFC、显著性和排序位置，是后续候选 TF、通路和药物重定位输入的基础。

相对路径：`results/ngs/GSE114012/tables/SW48/DEG/all_genes.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000160588	MPZL3	ENSG00000160588	196264	1.15525251121729	5.54272372439042	11.3002486691229	4.55074935566691e-07	0.00555727321123445	6.97532636507622
ENSG00000177076	ACER2	ENSG00000177076	340485	1.16736965420889	3.63446165297159	10.2569724493149	1.12957066881416e-06	0.00555727321123445	5.96620921958404
ENSG00000135679	MDM2	ENSG00000135679	4193	0.850117633129628	6.78840574810935	10.073849101689	1.33566893396117e-06	0.00555727321123445	5.92351351702526
ENSG00000183018	SPNS2	ENSG00000183018	124976	1.56926605665512	3.67970425392823	9.62577938388218	2.03490453114438e-06	0.00634991958943603	5.43078882719871
ENSG00000163235	TGFA	ENSG00000163235	7039	0.847463472680059	4.62680681951432	8.85925573782729	4.3462165053914e-06	0.0108498948840591	4.77155026648574
ENSG00000116106	EPHA4	ENSG00000116106	2043	0.867098040771775	4.97185155824257	8.33973505318967	7.49203930513423e-06	0.010909302934395	4.22595559593214
ENSG00000089472	HEPH	ENSG00000089472	9843	1.66528625473897	1.55352169552327	8.29427385222582	7.86743223183391e-06	0.010909302934395	3.71295319269992
ENSG000000281459	PADI1	ENSG000000281459	29943	3.40176235827948	-0.99692834285869	8.28440912436853	7.95155703901472e-06	0.010909302934395	2.52826520233191
ENSG00000134757	DSG3	ENSG00000134757	1830	1.69389037543511	2.46476890494012	8.06910301527646	1.00537955956984e-05	0.010909302934395	3.7870781089237
ENSG00000169247	SH3TC2	ENSG00000169247	79628	1.23359898186203	1.74606334953025	8.04865423852178	1.02828310226073e-05	0.010909302934395	3.59230237001372
ENSG00000128510	CPA4	ENSG00000128510	51200	3.71806846400913	1.25441997750113	8.01740012467375	1.06439017254158e-05	0.010909302934395	3.16586176001154
ENSG00000235750	KIAA0040	ENSG00000235750	9674	0.794824247455573	6.80777328743326	7.99057372068113	1.09648013720111e-05	0.010909302934395	3.79928640405713
ENSG00000135919	SERPINE2	ENSG00000135919	5270	0.848920238773747	4.43424448261457	7.95852176334424	1.1362036384164e-05	0.010909302934395	3.82830948894412
ENSG00000026103	FAS	ENSG00000026103	355	0.925264009744921	4.39227016460632	7.83207729847306	1.30883577058152e-05	0.0113494003749448	3.69003120145074
ENSG00000164938	TP53INP1	ENSG00000164938	94241	1.08165594925572	4.67566205756951	7.75592055436708	1.42639527307718e-05	0.0113494003749448	3.5952769901474
ENSG00000124762	CDKN1A	ENSG00000124762	1026	1.7526808001939	8.38370947968759	7.7385283581667	1.45481818618103e-05	0.0113494003749448	3.56263451027619
ENSG00000171617	ENC1	ENSG00000171617	8507	0.825311082252776	6.4756180652245	7.63039889211091	1.64589557960763e-05	0.0120847462498014	3.38633605131866
ENSG00000075223	SEMA3C	ENSG00000075223	10512	0.758658512558037	4.38703320482804	7.51302381077936	1.88455988363987e-05	0.012403554350754	3.32771502505776
ENSG00000156804	FBXO32	ENSG00000156804	114907	1.68128252174023	2.15161945390711	7.49493916860918	1.92455318655481e-05	0.012403554350754	3.1598404060936
ENSG00000133985	TTC9	ENSG00000133985	23508	0.866529154969597	4.42554696228995	7.46731376072323	1.98743059617914e-05	0.012403554350754	3.2737530702425
ENSG00000263309	KRT23	ENSG00000263309	25984	1.32645159291722	3.97074241106545	7.39427993235731	2.16462981526643e-05	0.0128661473115027	3.20783266932316

01. SW48: significant_genes

说明: 该表按统一阈值筛选显著差异基因, 代表当前模型中最直接的休眠样 LRC 转录特征, 并作为交集分析和火山图展示的核心输入。

相对路径: results/ngs/GSE114012/tables/SW48/DEG/significant_genes.csv

表格预览: 展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000160588	MPZL3	ENSG00000160588	196264	1.15525251121729	5.54272372439042	11.3002486691229	4.55074935566691e-07	0.00555727321123445	6.97532636507622
ENSG00000177076	ACER2	ENSG00000177076	340485	1.16736965420889	3.63446165297159	10.2569724493149	1.12957066881416e-06	0.00555727321123445	5.96620921958404
ENSG00000135679	MDM2	ENSG00000135679	4193	0.850117633129628	6.78840574810935	10.073849101689	1.33566893396117e-06	0.00555727321123445	5.92351351702526
ENSG00000183018	SPNS2	ENSG00000183018	124976	1.56926605665512	3.67970425392823	9.62577938388218	2.03490453114438e-06	0.00634991958943603	5.43078882719871
ENSG00000163235	TGFA	ENSG00000163235	7039	0.847463472680059	4.62680681951432	8.85925573782729	4.3462165053914e-06	0.0108498948840591	4.77155026648574
ENSG00000116106	EPHA4	ENSG00000116106	2043	0.867098040771775	4.97185155824257	8.33973505318967	7.49203930513423e-06	0.010909302934395	4.22595559593214
ENSG00000089472	HEPH	ENSG00000089472	9843	1.66528625473897	1.55352169552327	8.29427385222582	7.86743223183391e-06	0.010909302934395	3.71295319269992
ENSG00000281459	PADI1	ENSG00000281459	29943	3.40176235827948	-0.99692834285869	8.28440912436853	7.95155703901472e-06	0.010909302934395	2.52826520233191
ENSG00000134757	DSG3	ENSG00000134757	1830	1.69389037543511	2.46476890494012	8.06910301527646	1.00537955956984e-05	0.010909302934395	3.7870781089237
ENSG00000169247	SH3TC2	ENSG00000169247	79628	1.23359898186203	1.74606334953025	8.04865423852178	1.02828310226073e-05	0.010909302934395	3.59230237001372
ENSG00000128510	CPA4	ENSG00000128510	51200	3.71806846400913	1.25441997750113	8.01740012467375	1.06439017254158e-05	0.010909302934395	3.16586176001154
ENSG00000235750	KIAA0040	ENSG00000235750	9674	0.794824247455573	6.80777328743326	7.99057372068113	1.09648013720111e-05	0.010909302934395	3.79928640405713
ENSG00000135919	SERPINE2	ENSG00000135919	5270	0.848920238773747	4.43424448261457	7.95852176334424	1.1362036384164e-05	0.010909302934395	3.82830948894412
ENSG00000026103	FAS	ENSG00000026103	355	0.925264009744921	4.39227016460632	7.83207729847306	1.30883577058152e-05	0.0113494003749448	3.69003120145074
ENSG00000164938	TP53INP1	ENSG00000164938	94241	1.08165594925572	4.67566205756951	7.75592055436708	1.42639527307718e-05	0.0113494003749448	3.5952769901474
ENSG00000124762	CDKN1A	ENSG00000124762	1026	1.7526808001939	8.38370947968759	7.7385283581667	1.45481818618103e-05	0.0113494003749448	3.56263451027619
ENSG00000171617	ENC1	ENSG00000171617	8507	0.825311082252776	6.4756180652245	7.63039889211091	1.64589557960763e-05	0.0120847462498014	3.38633605131866
ENSG00000075223	SEMA3C	ENSG00000075223	10512	0.758658512558037	4.38703320482804	7.51302381077936	1.88455988363987e-05	0.012403554350754	3.32771502505776
ENSG00000156804	FBXO32	ENSG00000156804	114907	1.68128252174023	2.15161945390711	7.49493916860918	1.92455318655481e-05	0.012403554350754	3.1598404060936
ENSG00000133985	TTC9	ENSG00000133985	23508	0.866529154969597	4.42554696228995	7.46731376072323	1.98743059617914e-05	0.012403554350754	3.2737530702425
ENSG00000263309	KRT23	ENSG00000263309	25984	1.32645159291722	3.97074241106545	7.39427993235731	2.16462981526643e-05	0.0128661473115027	3.20783266932316

01. 差异分析 Summary: SW948

说明：该表概括当前 LRC vs BULK 比较的样本数、显著上调/下调基因数和阈值设置，用于判断每个细胞系或组合模型中休眠样 LRC 相对 BULK 的总体转录变化强度。

相对路径： `results/ngs/GSE114012/tables/SW948/DEG/summary.csv`

表格预览：展示全部列与前 21 行。

Dataset	Data_Type	Analysis_Name	Contrast	Samples_Used	Up	Down	Total_Significant_Genes	P_Value_Column	P_Value_Cutoff	LogFC_Cutoff	Microarray_Log2_Transformed
GSE114012	ngs	SW948	SW948_vs_BULK	8	305	57	362	P.Value	0.05	0.5	-

01. SW948: all_genes

说明：该表保留完整差异分析排序结果，可用于追溯每个基因在 LRC/BULK 模型中的 logFC、显著性和排序位置，是后续候选 TF、通路和药物重定位输入的基础。

相对路径： results/ngs/GSE114012/tables/SW948/DEG/all_genes.csv

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG000000002587	HS3ST1	ENSG000000002587	9957	-0.576817947077519	3.09205878117382	-4.993682785915	0.000274171983206565	0.241003896378655	0.0141147415431542
ENSG00000117000	RLF	ENSG00000117000	6018	0.664069794228499	5.06665486735482	4.98073437112463	0.000280368854062163	0.241003896378655	0.507026416059255
ENSG00000151414	NEK7	ENSG00000151414	140609	0.511946570029724	5.03732226478496	4.72375789234102	0.000438915903333984	0.241003896378655	0.136485358053095
ENSG00000187098	MITF	ENSG00000187098	4286	1.92213953840448	-0.00635373551924276	4.67937335202006	0.000474659634434272	0.241003896378655	-2.28916329933241
ENSG00000143751	SDE2	ENSG00000143751	163859	0.554894266011326	4.90668071788819	4.65926938902635	0.000491834755696976	0.241003896378655	0.0300817117339047
ENSG00000204950	LRRIC10B	ENSG00000204950	390205	-0.845735463025773	1.59222710262059	-4.56750785347248	0.000578844280381861	0.241003896378655	-1.28228871052374
ENSG00000165029	ABCA1	ENSG00000165029	19	0.977042825651498	2.31209939168595	4.55964634945018	0.000587007794261418	0.241003896378655	-0.893961549251028
ENSG00000163376	KBTBD8	ENSG00000163376	84541	0.715767646484417	2.29194946630444	4.53298279076451	0.00061559802720796	0.241003896378655	-0.912204897893217
ENSG00000144366	GULP1	ENSG00000144366	51454	0.614470550455211	4.2226885026875	4.51403274841831	0.000636794015803044	0.241003896378655	-0.267080867403667
ENSG00000105499	PLA2G4C	ENSG00000105499	8605	1.65515935050835	0.0526063263984802	4.49187144854953	0.000662546542513944	0.241003896378655	-2.33203813436879
ENSG00000155304	HSPA13	ENSG00000155304	6782	0.577271433162004	4.47555827771467	4.48372940409362	0.000672277554526319	0.241003896378655	-0.273198235457878
ENSG00000102934	PLLP	ENSG00000102934	51090	-0.425365023720661	3.96110417341938	-4.48155290125274	0.000674903861470372	0.241003896378655	-0.351978522696342
ENSG00000137710	RDX	ENSG00000137710	5962	0.622101594417767	4.337109469049	4.43827035134295	0.00072940238476853	0.241003896378655	-0.357787254117391
ENSG00000133134	BEX2	ENSG00000133134	84707	1.29760260999916	0.0535229897956448	4.42677704244213	0.000744627558254118	0.241003896378655	-2.34129623002472
ENSG00000147027	TMEM47	ENSG00000147027	83604	0.826152697486623	1.85225930835869	4.3958709054483	0.000787225340893701	0.241003896378655	-1.29979959298637
ENSG00000188994	ZNF292	ENSG00000188994	23036	0.480141316932004	6.10419443477969	4.33945124225339	0.000871624985756397	0.241003896378655	-0.391275404527264
ENSG00000164463	CREBRF	ENSG00000164463	153222	0.830731681327983	4.38994884062448	4.31330216128753	0.000913869031590849	0.241003896378655	-0.536827024771118
ENSG00000005249	PRKAR2B	ENSG00000005249	5577	-0.901989492362031	0.725017031596427	-4.22701514993002	0.00106892862361413	0.241003896378655	-2.06834869822079
ENSG00000166225	FRS2	ENSG00000166225	10818	0.543228643561158	4.93331134621017	4.22642648371577	0.00107007520015282	0.241003896378655	-0.61375181039846
ENSG00000198342	ZNF442	ENSG00000198342	79973	0.990378241738204	0.0723716394442486	4.21723192372744	0.00108814979039026	0.241003896378655	-2.43422258043248
ENSG00000118503	TNFAIP3	ENSG00000118503	7128	0.869091786656681	3.26121320940794	4.21067627004321	0.00110122922688067	0.241003896378655	-0.904814614062389

01. SW948: significant_genes

说明：该表按统一阈值筛选显著差异基因，代表当前模型中最直接的休眠样 LRC 转录特征，并作为交集分析和火山图展示的核心输入。

相对路径： results/ngs/GSE114012/tables/SW948/DEG/significant_genes.csv

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG000000002587	HS3ST1	ENSG000000002587	9957	-0.576817947077519	3.09205878117382	-4.993682785915	0.000274171983206565	0.241003896378655	0.0141147415431542
ENSG00000117000	RLF	ENSG00000117000	6018	0.664069794228499	5.06665486735482	4.98073437112463	0.000280368854062163	0.241003896378655	0.507026416059255
ENSG00000151414	NEK7	ENSG00000151414	140609	0.511946570029724	5.03732226478496	4.72375789234102	0.000438915903333984	0.241003896378655	0.136485358053095
ENSG00000187098	MITF	ENSG00000187098	4286	1.92213953840448	-0.00635373551924276	4.67937335202006	0.000474659634434272	0.241003896378655	-2.28916329933241
ENSG00000143751	SDE2	ENSG00000143751	163859	0.554894266011326	4.90668071788819	4.65926938902635	0.000491834755696976	0.241003896378655	0.0300817117339047
ENSG00000204950	LRRIC10B	ENSG00000204950	390205	-0.845735463025773	1.59222710262059	-4.56750785347248	0.000578844280381861	0.241003896378655	-1.28228871052374
ENSG00000165029	ABCA1	ENSG00000165029	19	0.977042825651498	2.31209939168595	4.55964634945018	0.000587007794261418	0.241003896378655	-0.893961549251028
ENSG00000163376	KBTBD8	ENSG00000163376	84541	0.715767646484417	2.29194946630444	4.53298279076451	0.00061559802720796	0.241003896378655	-0.912204897893217
ENSG00000144366	GULP1	ENSG00000144366	51454	0.614470550455211	4.2226885026875	4.51403274841831	0.000636794015803044	0.241003896378655	-0.267080867403667
ENSG00000105499	PLA2G4C	ENSG00000105499	8605	1.65515935050835	0.0526063263984802	4.49187144854953	0.000662546542513944	0.241003896378655	-2.33203813436879
ENSG00000155304	HSPA13	ENSG00000155304	6782	0.577271433162004	4.47555827771467	4.48372940409362	0.000672277554526319	0.241003896378655	-0.273198235457878
ENSG00000137710	RDX	ENSG00000137710	5962	0.622101594417767	4.337109469049	4.43827035134295	0.00072940238476853	0.241003896378655	-0.357787254117391
ENSG00000133134	BEX2	ENSG00000133134	84707	1.29760260999916	0.0535229897956448	4.42677704244213	0.000744627558254118	0.241003896378655	-2.34129623002472
ENSG00000147027	TMEM47	ENSG00000147027	83604	0.826152697486623	1.85225930835869	4.3958709054483	0.000787225340893701	0.241003896378655	-1.29979959298637
ENSG00000164463	CREBRF	ENSG00000164463	153222	0.830731681327983	4.38994884062448	4.31330216128753	0.000913869031590849	0.241003896378655	-0.536827024771118
ENSG00000005249	PRKAR2B	ENSG00000005249	5577	-0.901989492362031	0.725017031596427	-4.22701514993002	0.00106892862361413	0.241003896378655	-2.06834869822079
ENSG00000166225	FRS2	ENSG00000166225	10818	0.543228643561158	4.93331134621017	4.22642648371577	0.00107007520015282	0.241003896378655	-0.61375181039846
ENSG00000198342	ZNF442	ENSG00000198342	79973	0.990378241738204	0.0723716394442486	4.21723192372744	0.00108814979039026	0.241003896378655	-2.43422258043248
ENSG00000118503	TNFAIP3	ENSG00000118503	7128	0.869091786656681	3.26121320940794	4.21067627004321	0.00110122922688067	0.241003896378655	-0.904814614062389
ENSG00000162344	FGF19	ENSG00000162344	9965	1.2441252906983	0.247045526347607	4.16784103136048	0.00119078524981958	0.241003896378655	-2.39140039143566
ENSG00000204839	MROH6	ENSG00000204839	642475	-0.54688621687734	4.24969657797877	-4.14679600171064	0.0012375061261235	0.241003896378655	-0.788366548851609

02. 交集分析说明

- 输出目录: `results/ngs/GSE114012/intersect/<scheme>/`
- `summary.csv`: 每个交集方案的统计汇总 (交集基因数、共同上调、共同下调等)。
- `gene_list.csv`: 交集基因及其注释 (不含模型统计量列)。
- `deg_results.csv`: 交集基因在方案内每个参与分析中的完整 DEG 行信息。

02. DLD1/HCT15 交集统计

说明：该表汇总当前交集方案中跨细胞系重复出现的显著基因数量及共同上调/下调方向，用于优先筛选稳定的休眠维持或复苏相关候选信号。

相对路径：`results/ngs/GSE114012/intersect/DLD1_HCT15/summary.csv`

表格预览：展示全部列与前 21 行。

Selected_Analyses	DEG_Result_Analyses	Total_Intersected_Genes	Common_Up	Common_Down	Mixed_Direction
DLD1;HCT15	DLD1;HCT15	102	85	7	10

02. DLD1/HCT15 交集基因列表

说明：该表列出交集后的基因清单和基础注释，去除了模型统计量，便于将稳定候选基因继续输入 TF 富集、通路富集和药物重定位分析。

相对路径：`results/ngs/GSE114012/intersect/DLD1_HCT15/gene_list.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez
ENSG00000160179	ABCG1	ENSG00000160179	9619
ENSG00000168003	SLC3A2	ENSG00000168003	6520
ENSG00000051108	HERPUD1	ENSG00000051108	9709
ENSG00000165376	CLDN2	ENSG00000165376	9075
ENSG00000196878	LAMB3	ENSG00000196878	3914
ENSG00000187837	H1-2	ENSG00000187837	3006
ENSG00000006652	IFRD1	ENSG00000006652	3475
ENSG00000086548	CEACAM6	ENSG00000086548	4680
ENSG00000107738	VSIR	ENSG00000107738	64115
ENSG00000163584	RPL22L1	ENSG00000163584	200916
ENSG00000282937	PRSS22	ENSG00000282937	64063
ENSG00000185269	NOTUM	ENSG00000185269	147111
ENSG00000011028	MRC2	ENSG00000011028	9902
ENSG00000285069	SESN2	ENSG00000285069	83667
ENSG00000162772	ATF3	ENSG00000162772	467
ENSG00000174844	DNAH12	ENSG00000174844	201625
ENSG00000131016	AKAP12	ENSG00000131016	9590
ENSG00000136235	GNPMB	ENSG00000136235	10457
ENSG00000175197	DDIT3	ENSG00000175197	1649
ENSG00000176046	NUPR1	ENSG00000176046	26471
ENSG00000124107	SLPI	ENSG00000124107	6590

02. DLD1/HCT15 方案中的 DLD1 DEG 子集

说明：该表将交集基因回填到对应差异分析结果中，展示同一候选基因在单个细胞系 LRC vs BULK 比较中的方向、效应量和显著性。

相对路径： `results/ngs/GSE114012/intersect/DLD1_HCT15/DLD1/deg_results.csv`

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000160179	ABCG1	ENSG00000160179	9619	1.75263706690956	3.64968532444762	13.0816460461447	5.73435252410657e-08	0.000168104376284118	8.35328954100217
ENSG00000168003	SLC3A2	ENSG00000168003	6520	0.634594556602265	8.57470474823492	10.83115890642	3.8603838621999e-07	0.000378265047384138	7.08662254606513
ENSG00000051108	HERPUD1	ENSG00000051108	9709	0.5483753879604	6.7526859225706	10.7604032112199	4.12091114386612e-07	0.000378265047384138	7.0112032365038
ENSG00000165376	CLDN2	ENSG00000165376	9075	-1.40818237904057	4.6838561587445	-10.1256712740365	7.5278870889364e-07	0.000452553994920043	6.42256676585357
ENSG00000196878	LAMB3	ENSG00000196878	3914	0.703474021876742	8.73125570896915	10.0648790100083	7.98812795704189e-07	0.000452553994920043	6.35635874445822
ENSG00000187837	H1-2	ENSG00000187837	3006	1.15563280261529	5.40628502868344	9.29660250480426	1.73619865595687e-06	0.00058602479797548	5.61305277791167
ENSG00000006652	IFRD1	ENSG00000006652	3475	0.732343189402067	4.77066326146195	8.5672094896076	3.80959119448626e-06	0.000967187976200158	4.85276995972198
ENSG00000086548	CEACAM6	ENSG00000086548	4680	2.34847070572883	0.968671399724223	8.15211680721255	6.09815253492324e-06	0.00117371994967635	2.97182343053751
ENSG00000107738	VSIR	ENSG00000107738	64115	0.84784349347906	3.06485662996092	7.99200392141233	7.34672336986548e-06	0.00128547802963538	4.11476159298788
ENSG00000163584	RPL22L1	ENSG00000163584	200916	-0.50205366702381	7.56991494185644	-7.64483626744416	1.11085626786531e-05	0.00153141320294588	3.61854146082673
ENSG00000282937	PRSS22	ENSG00000282937	64063	0.821511584150494	4.77458763576748	7.37432176398783	1.54763375318108e-05	0.00169489885756851	3.45656120584568
ENSG00000185269	NOTUM	ENSG00000185269	147111	0.648805683269967	3.38281165316353	7.35838232707222	1.57858791716732e-05	0.00169489885756851	3.45911921503259
ENSG00000011028	MRC2	ENSG00000011028	9902	2.0652640668065	-0.608248977460979	7.32074125404598	1.6543625196763e-05	0.00169646461096401	1.35914795107387
ENSG00000285069	SESN2	ENSG00000285069	83667	0.716461332298402	4.32221096817625	7.30982008578172	1.67707344920755e-05	0.00169646461096401	3.40564662711083
ENSG00000162772	ATF3	ENSG00000162772	467	1.81464980017736	2.62620060872411	7.21651958865385	1.88534188536595e-05	0.00176690680642426	3.13963845879164
ENSG00000174844	DNAH12	ENSG00000174844	201625	2.07967632447134	-0.47981902972442	7.04905163777415	2.33228418838337e-05	0.00191129213108784	1.23316734872172
ENSG00000131016	AKAP12	ENSG00000131016	9590	0.57294773430711	4.46751530295746	7.00069437895826	2.48162517830176e-05	0.00196788421782995	3.00649291945773
ENSG00000136235	GPNUMB	ENSG00000136235	10457	2.3689999963388	-0.560688697701015	6.90644092211835	2.80307189417411e-05	0.00211625381901947	1.15810676784743
ENSG00000175197	DDIT3	ENSG00000175197	1649	1.03302707399971	3.91419399033181	6.16864576842681	7.56408172342238e-05	0.00348539964963961	1.94622361793009
ENSG00000176046	NUPR1	ENSG00000176046	26471	1.62455057669399	0.399507865854899	6.15379847464226	7.72239832477043e-05	0.00349743526650695	1.08932250332548
ENSG00000124107	SLPI	ENSG00000124107	6590	0.63635665708428	2.99999902306739	6.129469923217	7.98952410227811e-05	0.0035919568776492	1.91639058908962

02. DLD1/HCT15 方案中的 HCT15 DEG 子集

说明：该表将交集基因回填到对应差异分析结果中，展示同一候选基因在单个细胞系 LRC vs BULK 比较中的方向、效应量和显著性。

相对路径：`results/ngs/GSE114012/intersect/DLD1_HCT15/HCT15/deg_results.csv`

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000160179	ABCG1	ENSG00000160179	9619	0.787246767807064	2.59644869641349	4.25262699847098	0.00155461781415425	0.0824032546552309	-0.833578373501279
ENSG00000168003	SLC3A2	ENSG00000168003	6520	0.803425954568501	10.1059620250871	2.88308185788107	0.0157561719414558	0.182613894468094	-3.36630066065941
ENSG00000051108	HERPUD1	ENSG00000051108	9709	0.577493919272912	7.47010114164978	4.51327437433573	0.00102492663148639	0.0768703863137253	-0.7261702644757771
ENSG00000165376	CLDN2	ENSG00000165376	9075	-0.724875786282873	6.56645320369239	-4.41001365113524	0.00120751002681934	0.078034577628331	-0.890774093931572
ENSG00000196878	LAMB3	ENSG00000196878	3914	0.583785326581605	8.15827831492715	5.30218319928334	0.000307831703079213	0.0542989182733616	0.506514858453075
ENSG00000187837	H1-2	ENSG00000187837	3006	0.654515130894633	6.47576810982993	3.43420997124469	0.00607745053849332	0.128571193202965	-2.51923452136343
ENSG00000006652	IFRD1	ENSG00000006652	3475	0.705220793135199	5.29190717830141	2.57531579754383	0.0269348723623087	0.214104547083927	-3.89340886429779
ENSG00000086548	CEACAM6	ENSG00000086548	4680	1.11913866216605	0.688335375254393	5.29589597342006	0.000310687666135637	0.0542989182733616	0.227895298428174
ENSG00000107738	VSIR	ENSG00000107738	64115	1.13802569042159	2.3662688518295	6.74296375525147	4.28334367584577e-05	0.0332614234499059	2.2961955914556
ENSG00000163584	RPL22L1	ENSG00000163584	200916	-0.55530948411516	6.47068001645389	-3.78316983123518	0.00336662009149431	0.104571180771333	-1.92746945729334
ENSG00000282937	PRSS22	ENSG00000282937	64063	0.691329511093788	4.58058437194155	3.96669771293531	0.00248083318101415	0.0935699394930508	-1.46255163587065
ENSG00000185269	NOTUM	ENSG00000185269	147111	-0.755454362047889	6.33710146169876	-3.38919554660183	0.00656419440831591	0.13169290331942	-2.59080974381456
ENSG00000011028	MRC2	ENSG00000011028	9902	-0.560914231363555	6.05852674325117	-4.45890970707778	0.0011171139215735	0.0768703863137253	-0.795202813274244
ENSG00000285069	SESN2	ENSG00000285069	83667	0.813650393370951	5.27568994566573	2.95274056901514	0.0139570844482938	0.174807848009418	-3.25055264602123
ENSG00000162772	ATF3	ENSG00000162772	467	1.24338585742293	4.2450824266744	6.51804111509626	5.72245266635181e-05	0.0347154177656026	2.26133374160009
ENSG00000174844	DNAH12	ENSG00000174844	201625	0.759774824573755	1.20290108621336	3.71459944216823	0.00377717858844826	0.109951994327971	-1.63456426570165
ENSG00000131016	AKAP12	ENSG00000131016	9590	0.693954618761413	5.62319339692421	2.62031437324008	0.0249054412974387	0.209027351061038	-3.85431094754777
ENSG00000136235	GPNUMB	ENSG00000136235	10457	1.53327466559777	1.4367056494721	6.32014155195132	7.42342369286293e-05	0.0347154177656026	1.52992381468811
ENSG00000175197	DDIT3	ENSG00000175197	1649	1.25173513042871	5.0066454846129	3.62644751836073	0.00438265278033721	0.1161755007093	-2.0663110105679
ENSG00000176046	NUPR1	ENSG00000176046	26471	1.2524638488618	0.372879908355705	3.5844201269373	0.00470592375889779	0.118330489556288	-1.88415974731524
ENSG00000124107	SLPI	ENSG00000124107	6590	1.80181890412888	3.36532875338617	7.91840728066626	1.04068541066609e-05	0.0230353946901153	3.7050457244412

02. DLD1/HCT15/SW48 交集统计

说明：该表汇总当前交集方案中跨细胞系重复出现的显著基因数量及共同上调/下调方向，用于优先筛选稳定的休眠维持或复苏相关候选信号。

相对路径： `results/ngs/GSE114012/intersect/DLD1_HCT15_SW48/summary.csv`

表格预览：展示全部列与前 21 行。

Selected_Analyses		DEG_Result_Analyses	Total_Intersected_Genes	Common_Up	Common_Down	Mixed_Direction
DLD1;HCT15;SW48	DLD1;HCT15;SW48		19	17	1	1

02. DLD1/HCT15/SW48 交集基因列表

说明：该表列出交集后的基因清单和基础注释，去除了模型统计量，便于将稳定候选基因继续输入 TF 富集、通路富集和药物重定位分析。

相对路径：`results/ngs/GSE114012/intersect/DLD1_HCT15_SW48/gene_list.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez
ENSG00000086548	CEACAM6	ENSG00000086548	4680
ENSG00000282937	PRSS22	ENSG00000282937	64063
ENSG00000162772	ATF3	ENSG00000162772	467
ENSG00000131016	AKAP12	ENSG00000131016	9590
ENSG00000124107	SLPI	ENSG00000124107	6590
ENSG00000120885	CLU	ENSG00000120885	1191
ENSG00000078081	LAMP3	ENSG00000078081	27074
ENSG00000183091	NEB	ENSG00000183091	4703
ENSG00000087589	CASS4	ENSG00000087589	57091
ENSG00000196208	GREB1	ENSG00000196208	9687
ENSG00000057657	PRDM1	ENSG00000057657	639
ENSG00000129226	CD68	ENSG00000129226	968
ENSG00000158406	H4C8	ENSG00000158406	8365
ENSG00000174885	NLRP6	ENSG00000174885	171389
ENSG00000162733	DDR2	ENSG00000162733	4921
ENSG00000204421	LY6G6C	ENSG00000204421	80740
ENSG00000162496	DHRS3	ENSG00000162496	9249
ENSG00000229341	TNXB	ENSG00000229341	7148
ENSG00000205426	KRT81	ENSG00000205426	3887

02. DLD1/HCT15/SW48 方案中的 DLD1 DEG 子集

说明：该表将交集基因回填到对应差异分析结果中，展示同一候选基因在单个细胞系 LRC vs BULK 比较中的方向、效应量和显著性。

相对路径： `results/ngs/GSE114012/intersect/DLD1_HCT15_SW48/DLD1/deg_results.csv`

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000086548	CEACAM6	ENSG00000086548	4680	2.34847070572883	0.968671399724223	8.15211680721255	6.09815253492324e-06	0.00117371994967635	2.97182343053751
ENSG00000282937	PRSS22	ENSG00000282937	64063	0.821511584150494	4.77458763576748	7.37432176398783	1.54763375318108e-05	0.00169489885756851	3.45656120584568
ENSG00000162772	ATF3	ENSG00000162772	467	1.81464980017736	2.62620060872411	7.21651958865385	1.88534188536595e-05	0.00176690680642426	3.13963845879164
ENSG00000131016	AKAP12	ENSG00000131016	9590	0.57294773430711	4.46751530295746	7.00069437895826	2.48162517830176e-05	0.00196788421782995	3.00649291945773
ENSG00000124107	SLPI	ENSG00000124107	6590	0.63635665708428	2.99999902306739	6.129469923217	7.98952410227811e-05	0.0035919568776492	1.91639058908962
ENSG00000120885	CLU	ENSG00000120885	1191	0.809558285147618	7.66044888101851	5.68234819643668	0.000151436578626552	0.00504061907469563	0.903130423608865
ENSG00000078081	LAMP3	ENSG00000078081	27074	1.00665907485866	2.35117765127651	4.92735450459784	0.000475194645545318	0.00938780136538012	0.239538404759207
ENSG00000183091	NEB	ENSG00000183091	4703	1.01602080534454	2.46107180905745	4.78706491932586	0.000592961231911758	0.0104457986813516	0.0348163931585557
ENSG00000087589	CASS4	ENSG00000087589	57091	1.14943378338692	0.291052667920189	4.29387343810483	0.00131964175044536	0.0163396250075044	-0.917355839666105
ENSG00000196208	GREB1	ENSG00000196208	9687	1.50845594898728	-0.474689632328964	4.14891328938825	0.0016797683643875	0.0191038525547956	-1.30759952362561
ENSG00000057657	PRDM1	ENSG00000057657	639	0.653847485342044	1.63255572830062	3.38329076830929	0.00625243131745712	0.041547181260075	-2.09082805577223
ENSG00000129226	CD68	ENSG00000129226	968	1.42521178045985	2.14715877465073	3.12491835845379	0.00985604413085118	0.055509377732171	-2.54526738743638
ENSG00000158406	H4C8	ENSG00000158406	8365	1.16979635195528	-0.0785359303593899	3.12023321598541	0.00993805588168349	0.0557985149460624	-2.50179027638352
ENSG00000174885	NLRP6	ENSG00000174885	171389	-0.615734760081328	1.73684630970156	-3.07675093418886	0.0107329886324227	0.0587631347722725	-2.58532639396756
ENSG00000162733	DDR2	ENSG00000162733	4921	0.637483022587211	0.219383770708485	2.47403284893153	0.0312420099562115	0.113202046334503	-3.38446304702835
ENSG00000204421	LY6G6C	ENSG00000204421	80740	0.537511775755852	2.38682411248374	2.37135920956291	0.0374311893416758	0.127575425005533	-3.83679801245469
ENSG00000162496	DHR53	ENSG00000162496	9249	0.599369581763371	2.06347166671245	2.35927728607318	0.0382335860286234	0.129390609487354	-3.78708318909398
ENSG00000229341	TNXB	ENSG00000229341	7148	0.607789473285428	2.21527469602656	2.35399780046058	0.0385894262427657	0.130254403282412	-3.82661171873073
ENSG00000205426	KRT81	ENSG00000205426	3887	0.635387443958113	-0.0982696749400837	2.22827635174195	0.0480729828517783	0.15027739786693	-3.70515233311944

02. DLD1/HCT15/SW48 方案中的 HCT15 DEG 子集

说明：该表将交集基因回填到对应差异分析结果中，展示同一候选基因在单个细胞系 LRC vs BULK 比较中的方向、效应量和显著性。

相对路径： `results/ngs/GSE114012/intersect/DLD1_HCT15_SW48/HCT15/deg_results.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000086548	CEACAM6	ENSG00000086548	4680	1.11913866216605	0.688335375254393	5.29589597342006	0.000310687666135637	0.0542989182733616	0.227895298428174
ENSG00000282937	PRSS22	ENSG00000282937	64063	0.691329511093788	4.58058437194155	3.96669771293531	0.00248083318101415	0.0935699394930508	-1.46255163587065
ENSG00000162772	ATF3	ENSG00000162772	467	1.24338585742293	4.2450824266744	6.51804111509626	5.72245266635181e-05	0.0347154177656026	2.26133374160009
ENSG00000131016	AKAP12	ENSG00000131016	9590	0.693954618761413	5.62319339692421	2.62031437324008	0.0249054412974387	0.209027351061038	-3.85431094754777
ENSG00000124107	SLPI	ENSG00000124107	6590	1.80181890412888	3.36532875338617	7.91840728066626	1.04068541066609e-05	0.0230353946901153	3.7050457244412
ENSG00000120885	CLU	ENSG00000120885	1191	0.831312399671628	7.19696729314522	6.17543107879907	9.00875120603811e-05	0.0371639139596591	1.73904515740864
ENSG00000078081	LAMP3	ENSG00000078081	27074	1.13324013832028	3.73184673913899	6.4662389510687	6.12284391223567e-05	0.0347154177656026	2.18594254887515
ENSG00000183091	NEB	ENSG00000183091	4703	0.844919935210186	1.73654003609668	3.34863602827879	0.00703709035370411	0.135615517896712	-2.17082272578744
ENSG00000087589	CASS4	ENSG00000087589	57091	1.41117737432089	0.568560232538333	5.59830789454786	0.000200527661270554	0.0457878846213985	0.478927197563364
ENSG00000196208	GREB1	ENSG00000196208	9687	0.630897537510179	1.50237417929089	2.91977663312451	0.014781056323418	0.178150852719502	-2.81093755207461
ENSG00000057657	PRDM1	ENSG00000057657	639	1.48030431496762	0.75591379186523	3.25910430610956	0.00820881379186979	0.14333935299798	-2.30000485640275
ENSG00000129226	CD68	ENSG00000129226	968	1.7296999742701	3.45544611769598	4.14253088109799	0.0018588201151046	0.0876864212060263	-1.04147416797467
ENSG00000158406	H4C8	ENSG00000158406	8365	0.786681496348874	1.32603750558737	3.89538710457637	0.00279198350898201	0.0957259998588934	-1.37275551512786
ENSG00000174885	NLRP6	ENSG00000174885	171389	-0.80795770500189	1.12509621418863	-3.04646924355414	0.0118591784660072	0.166900868795055	-2.60319360475777
ENSG00000162733	DDR2	ENSG00000162733	4921	0.622241710947171	0.520126125113129	2.41623685180621	0.0355060500372309	0.239997627517401	-3.48735242722056
ENSG00000204421	LY6G6C	ENSG00000204421	80740	0.547982609148633	2.02842905696843	2.96305161841532	0.0137089971585123	0.173854860663679	-2.78981170882534
ENSG00000162496	DHRS3	ENSG00000162496	9249	1.05051864022264	0.861542377442147	4.04083278600155	0.00219551634469403	0.0894245791503464	-1.22193259392276
ENSG00000229341	TNXB	ENSG00000229341	7148	0.63301862696263	2.65365905753426	2.83132225609802	0.0172427886631262	0.18650533615666	-3.07481179321633
ENSG00000205426	KRT81	ENSG00000205426	3887	0.888954750957592	-0.3992230137654	2.76879068726773	0.0192279815965021	0.195057185075994	-3.00879345204485

02. DLD1/HCT15/SW48 方案中的 SW48 DEG 子集

说明：该表将交集基因回填到对应差异分析结果中，展示同一候选基因在单个细胞系 LRC vs BULK 比较中的方向、效应量和显著性。

相对路径：`results/ngs/GSE114012/intersect/DLD1_HCT15_SW48/SW48/deg_results.csv`

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000086548	CEACAM6	ENSG00000086548	4680	1.83439804658535	0.227802346191551	2.65996780808211	0.0236475826133733	0.278987831928285	-3.19935079439718
ENSG00000282937	PRSS22	ENSG00000282937	64063	1.51615552391205	1.78393116877101	6.88597406487023	3.99006585144359e-05	0.0166013339859063	2.46469166075655
ENSG00000162772	ATF3	ENSG00000162772	467	1.37095467194176	3.04753470366309	2.90842432378457	0.0154022092306372	0.242740373253553	-3.17942942126081
ENSG00000131016	AKAP12	ENSG00000131016	9590	0.761379852198632	7.03353477266767	4.8866585663376	0.000611341608008563	0.0563407342726756	-0.343478920376727
ENSG00000124107	SLPI	ENSG00000124107	6590	1.3288107702217	3.14165232393575	3.01304094049803	0.0128638837874741	0.225515445836028	-3.02281647587112
ENSG00000120885	CLU	ENSG00000120885	1191	0.891940553322362	7.73244834026536	5.11178460340315	0.00043729324577858	0.0483034893257366	0.040508850877611
ENSG00000078081	LAMP3	ENSG00000078081	27074	-0.649508974756801	3.63401268280863	-2.76125393337899	0.019853790068644	0.266465341079402	-3.54570227222087
ENSG00000183091	NEB	ENSG00000183091	4703	0.870908370116398	0.868158602329421	2.67650111536213	0.0229821633425079	0.27718682853977	-3.23940034471064
ENSG00000087589	CASS4	ENSG00000087589	57091	1.02840897835661	0.208858525173881	3.79504687276858	0.00343124388016558	0.127847122723065	-1.53794296305247
ENSG00000196208	GREB1	ENSG00000196208	9687	1.89745064466209	-0.946363934215299	4.39522047116596	0.00130266143635949	0.0883685872208649	-0.885226655618905
ENSG00000057657	PRDM1	ENSG00000057657	639	0.678315976633445	6.29622015793876	2.49555874613576	0.0314002291981715	0.307919833178557	-4.32975026529036
ENSG00000129226	CD68	ENSG00000129226	968	0.613724970395683	5.75122256346463	3.10586879833915	0.0109687280737238	0.212039477844021	-3.26057917434659
ENSG00000158406	H4C8	ENSG00000158406	8365	0.827930906084546	-0.0120208775098356	3.42596439055837	0.00635903563790063	0.168626271044201	-2.06850240242386
ENSG00000174885	NLRP6	ENSG00000174885	171389	-0.740949200793741	0.551101559908127	-2.4873351078581	0.0318480622595833	0.308181741274627	-3.49367885167513
ENSG00000162733	DDR2	ENSG00000162733	4921	0.843314379880288	0.417578690391784	2.88472983301296	0.0160442852028251	0.24542250968341	-2.88191653635614
ENSG00000204421	LY6G6C	ENSG00000204421	80740	0.922208494345584	3.45704208693838	5.11334848197156	0.000436287721682316	0.0483034893257366	0.269089716373916
ENSG00000162496	DHR53	ENSG00000162496	9249	0.722277264793861	1.62605112101464	2.59107977452695	0.0266329729629649	0.290176544495433	-3.47238500821661
ENSG00000229341	TNXB	ENSG00000229341	7148	0.849676473609322	1.26083835608492	3.10956947506763	0.0108993634149087	0.212039477844021	-2.60373269160474
ENSG00000205426	KRT81	ENSG00000205426	3887	1.21032375656766	3.88740729667532	2.69274842623984	0.0223464617481332	0.273996596797838	-3.69931335654608

02. HT55/SW948 交集统计

说明：该表汇总当前交集方案中跨细胞系重复出现的显著基因数量及共同上调/下调方向，用于优先筛选稳定的休眠维持或复苏相关候选信号。

相对路径： `results/ngs/GSE114012/intersect/HT55_SW948/summary.csv`

表格预览：展示全部列与前 21 行。

Selected_Analyses	DEG_Result_Analyses	Total_Intersected_Genes	Common_Up	Common_Down	Mixed_Direction
HT55;SW948	HT55;SW948	49	43	2	4

02. HT55/SW948 交集基因列表

说明：该表列出交集后的基因清单和基础注释，去除了模型统计量，便于将稳定候选基因继续输入 TF 富集、通路富集和药物重定位分析。

相对路径：`results/ngs/GSE114012/intersect/HT55_SW948/gene_list.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez
ENSG00000059804	SLC2A3	ENSG00000059804	6515
ENSG00000285069	SESN2	ENSG00000285069	83667
ENSG00000140853	NLRC5	ENSG00000140853	84166
ENSG00000153233	PTPRR	ENSG00000153233	5801
ENSG00000198832	SELENOM	ENSG00000198832	140606
ENSG00000175197	DDIT3	ENSG00000175197	1649
ENSG00000184678	H2BC21	ENSG00000184678	8349
ENSG00000132823	OSER1	ENSG00000132823	51526
ENSG00000179094	PER1	ENSG00000179094	5187
ENSG00000184613	NELL2	ENSG00000184613	4753
ENSG00000288859	H2AC19	ENSG00000288859	723790
ENSG00000288825	H2AC18	ENSG00000288825	8337
ENSG00000126368	NR1D1	ENSG00000126368	9572
ENSG00000229341	TNXB	ENSG00000229341	7148
ENSG00000100292	HMOX1	ENSG00000100292	3162
ENSG00000122884	P4HA1	ENSG00000122884	5033
ENSG00000280501	CEACAM7	ENSG00000280501	1087
ENSG00000078804	TP53INP2	ENSG00000078804	58476
ENSG00000160179	ABCG1	ENSG00000160179	9619
ENSG00000211445	GPX3	ENSG00000211445	2878
ENSG00000152952	PLOD2	ENSG00000152952	5352

02. HT55/SW948 方案中的 HT55 DEG 子集

说明：该表将交集基因回填到对应差异分析结果中，展示同一候选基因在单个细胞系 LRC vs BULK 比较中的方向、效应量和显著性。

相对路径：`results/ngs/GSE114012/intersect/HT55_SW948/HT55/deg_results.csv`

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000059804	SLC2A3	ENSG00000059804	6515	1.66518145583407	6.80095289108856	5.3500537469641	0.000222160764810581	0.405693176761372	0.884403744862246
ENSG00000285069	SESN2	ENSG00000285069	83667	0.940276054652189	2.49812737530616	5.16197501253512	0.000297759188862012	0.405693176761372	0.00564894212721434
ENSG00000140853	NLR5	ENSG00000140853	84166	-0.58598522055533	4.61462501719449	-4.93736731678053	0.000425223859328897	0.405693176761372	0.263686850628111
ENSG00000153233	PTPRR	ENSG00000153233	5801	0.780626885373714	4.48602828209368	4.82931375069941	0.000506015140450154	0.405693176761372	0.102447894678306
ENSG00000198832	SELENOM	ENSG00000198832	140606	1.99704548561285	-1.20618961278681	4.80596390890353	0.000525510771443461	0.405693176761372	-2.13919951409128
ENSG00000175197	DDIT3	ENSG00000175197	1649	0.801262967544231	3.97870901402782	4.74983731812166	0.000575679463335577	0.405693176761372	-0.0579139788854031
ENSG00000184678	H2BC21	ENSG00000184678	8349	0.87021548941066	6.30815595885261	4.29767483829708	0.00121889663442434	0.413218410046681	-0.637427224298803
ENSG00000132823	OSER1	ENSG00000132823	51526	0.52909268313249	5.5425980688796	4.27855655840796	0.00125892270940136	0.413218410046681	-0.666911307543808
ENSG00000179094	PER1	ENSG00000179094	5187	0.662511131893083	4.09510754855917	4.26276440137768	0.00129301975845976	0.413218410046681	-0.724174884552938
ENSG00000184613	NELL2	ENSG00000184613	4753	0.744642226719339	5.08115181216889	4.20246161166096	0.00143234767928068	0.413218410046681	-0.784874121491482
ENSG00000288859	H2AC19	ENSG00000288859	723790	1.00229637171469	4.31776513346619	4.19911476262525	0.00144052514923439	0.413218410046681	-0.807991996189171
ENSG00000288825	H2AC18	ENSG00000288825	8337	0.997375803328962	4.31927812969034	4.19340443258852	0.00145458996878193	0.413218410046681	-0.816259499951535
ENSG00000126368	NR1D1	ENSG00000126368	9572	0.701054952814586	6.38845861628439	3.99395904680035	0.00204736834261706	0.413218410046681	-1.11112088590526
ENSG00000229341	TNXB	ENSG00000229341	7148	0.997362688534974	0.0105715325944354	3.96455216639767	0.00215404955478164	0.413218410046681	-2.27366368513482
ENSG00000100292	HMOX1	ENSG00000100292	3162	0.658084928082908	3.06501280383202	3.96413182556204	0.00215561560444849	0.413218410046681	-1.25416760814022
ENSG00000122884	P4HA1	ENSG00000122884	5033	0.67235888849352	6.41407039881537	3.96149541890934	0.00216546495621577	0.413218410046681	-1.16260064612741
ENSG00000280501	CEACAM7	ENSG00000280501	1087	1.93121702867123	1.78879965944844	3.95158190768148	0.00220292069897237	0.413218410046681	-1.63985864491978
ENSG00000078804	TP53INP2	ENSG00000078804	58476	0.76247629510337	5.44489913471784	3.84496878813947	0.00265088378504874	0.422031170304092	-1.33948380425532
ENSG00000160179	ABCG1	ENSG00000160179	9619	0.619699449900384	4.72320955269797	3.69284733238397	0.00345932057471079	0.440505279862276	-1.57084646300525
ENSG00000211445	GPX3	ENSG00000211445	2878	0.938105330391414	3.78043683320218	3.58734031406557	0.0041661440125156	0.449671647865222	-1.73589745970633
ENSG00000152952	PLOD2	ENSG00000152952	5352	0.834316588603373	4.18300701955419	3.55409565227389	0.00441839637577683	0.463722730552341	-1.78204845335163

02. HT55/SW948 方案中的 SW948 DEG 子集

说明：该表将交集基因回填到对应差异分析结果中，展示同一候选基因在单个细胞系 LRC vs BULK 比较中的方向、效应量和显著性。

相对路径：`results/ngs/GSE114012/intersect/HT55_SW948/SW948/deg_results.csv`

表格预览：展示全部列与前 21 行。

GenelD	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000059804	SLC2A3	ENSG00000059804	6515	3.00887274773639	0.334513584718167	2.9816034440168	0.0109807278950822	0.241003896378655	-3.30696238962225
ENSG00000285069	SESN2	ENSG00000285069	83667	0.784398243217646	4.20909235336918	3.37808669718944	0.00518425769583146	0.241003896378655	-1.96294061320073
ENSG00000140853	NLR5	ENSG00000140853	84166	-0.585503366496002	2.88636778537025	-2.84674416568823	0.0141715698943653	0.241003896378655	-2.81872811606499
ENSG00000153233	PTPRR	ENSG00000153233	5801	0.572465502133387	5.46646974713958	2.38848099961975	0.0334271662859984	0.272785334103748	-3.56897934738621
ENSG00000198832	SELENOM	ENSG00000198832	140606	1.46701895265241	0.324038122551013	2.7545996965003	0.0168629477241586	0.242821735665431	-3.41223688435722
ENSG00000175197	DDIT3	ENSG00000175197	1649	1.73762051486941	4.90996641569133	3.37703390625466	0.00519457660153073	0.241003896378655	-1.95411704832623
ENSG00000184678	H2BC21	ENSG00000184678	8349	0.592448990489238	3.47269824915055	2.606275333207	0.0222841560064526	0.251902755627717	-3.14253653919849
ENSG00000132823	OSER1	ENSG00000132823	51526	0.599624011800612	6.25217900690758	3.7963700515941	0.00236403906930412	0.241003896378655	-1.26941707673246
ENSG00000179094	PER1	ENSG00000179094	5187	0.537948767710982	4.80898417592448	2.62432805930838	0.021542781185271	0.24942062025565	-3.16173800309203
ENSG00000184613	NELL2	ENSG00000184613	4753	0.64711021503916	2.10604299909151	2.47239491784681	0.0286110462961629	0.261950392146382	-3.37362872975201
ENSG00000288859	H2AC19	ENSG00000288859	723790	0.730335177951955	1.96979897336377	2.45054263144963	0.0297966003481518	0.265228813865053	-3.41304835619504
ENSG00000288825	H2AC18	ENSG00000288825	8337	0.734014033709503	1.97173809828519	2.45650185363288	0.0294686784278319	0.264132741043903	-3.40587413223377
ENSG00000126368	NR1D1	ENSG00000126368	9572	1.32829774564094	5.03583347681531	3.33774335864114	0.00559489372478073	0.241003896378655	-2.01643703608483
ENSG00000229341	TNXB	ENSG00000229341	7148	0.704110941821204	1.81884866552728	2.43937098831048	0.0304208469219603	0.266012260067419	-3.44044216576865
ENSG00000100292	HMOX1	ENSG00000100292	3162	0.667699265654353	3.1438333555162	2.88192429619445	0.0132599524244151	0.241003896378655	-2.7538582402596
ENSG00000122884	P4HA1	ENSG00000122884	5033	0.563603336512751	6.5434607621884	3.03806929667424	0.00986711742104504	0.241003896378655	-2.5475519961095
ENSG00000280501	CEACAM7	ENSG00000280501	1087	1.11847288779713	1.43380821555976	2.55393981398741	0.0245765800150712	0.254201065785395	-3.37095369900913
ENSG00000078804	TP53INP2	ENSG00000078804	58476	1.18794913418422	3.22667352454897	3.48624532716573	0.00422744374551357	0.241003896378655	-1.90298346957735
ENSG00000160179	ABCG1	ENSG00000160179	9619	0.993563244923847	2.43074930322327	2.89290119724634	0.0129875719273012	0.241003896378655	-2.81750954568906
ENSG00000211445	GPX3	ENSG00000211445	2878	0.722434171267149	4.86230302702555	2.73273704335972	0.017571979756095	0.242937243615241	-2.99045373525692
ENSG00000152952	PLOD2	ENSG00000152952	5352	0.605902512144281	6.3560044010132	3.01720013601352	0.0102649417879696	0.241003896378655	-2.57711838651925

02. SW948/RKO 交集统计

说明：该表汇总当前交集方案中跨细胞系重复出现的显著基因数量及共同上调/下调方向，用于优先筛选稳定的休眠维持或复苏相关候选信号。

相对路径： `results/ngs/GSE114012/intersect/SW948_RKO/summary.csv`

表格预览：展示全部列与前 21 行。

Selected_Analyses	DEG_Result_Analyses	Total_Intersected_Genes	Common_Up	Common_Down	Mixed_Direction
SW948;RKO	SW948;RKO	83	68	1	14

02. SW948/RKO 交集基因列表

说明：该表列出交集后的基因清单和基础注释，去除了模型统计量，便于将稳定候选基因继续输入 TF 富集、通路富集和药物重定位分析。

相对路径：`results/ngs/GSE114012/intersect/SW948_RKO/gene_list.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez
ENSG00000187098	MITF	ENSG00000187098	4286
ENSG00000165029	ABCA1	ENSG00000165029	19
ENSG00000118503	TNFAIP3	ENSG00000118503	7128
ENSG00000204839	MROH6	ENSG00000204839	642475
ENSG00000123095	BHLHE41	ENSG00000123095	79365
ENSG00000113742	CPEB4	ENSG00000113742	80315
ENSG00000153094	BCL2L11	ENSG00000153094	10018
ENSG00000144824	PHLDB2	ENSG00000144824	90102
ENSG00000142871	CCN1	ENSG00000142871	3491
ENSG00000078081	LAMP3	ENSG00000078081	27074
ENSG00000139112	GABARAPL1	ENSG00000139112	23710
ENSG00000146411	SLC2A12	ENSG00000146411	154091
ENSG00000019485	PRDM11	ENSG00000019485	56981
ENSG00000104081	BMF	ENSG00000104081	90427
ENSG00000105290	APLP1	ENSG00000105290	333
ENSG00000105974	CAV1	ENSG00000105974	857
ENSG00000120885	CLU	ENSG00000120885	1191
ENSG00000184205	TSPYL2	ENSG00000184205	64061
ENSG00000162645	GBP2	ENSG00000162645	2634
ENSG00000128590	DNAJB9	ENSG00000128590	4189
ENSG00000151117	TMEM86A	ENSG00000151117	144110

02. SW948/RKO 方案中的 SW948 DEG 子集

说明：该表将交集基因回填到对应差异分析结果中，展示同一候选基因在单个细胞系 LRC vs BULK 比较中的方向、效应量和显著性。

相对路径： results/ngs/GSE114012/intersect/SW948_RKO/SW948/deg_results.csv

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000187098	MITF	ENSG00000187098	4286	1.92213953840448	-0.00635373551924276	4.67937335202006	0.000474659634434272	0.241003896378655	-2.28916329933241
ENSG00000165029	ABCA1	ENSG00000165029	19	0.977042825651498	2.31209939168595	4.55964634945018	0.000587007794261418	0.241003896378655	-0.893961549251028
ENSG00000118503	TNFAIP3	ENSG00000118503	7128	0.869091786656681	3.26121320940794	4.21067627004321	0.00110122922688067	0.241003896378655	-0.904814614062389
ENSG00000204839	MROH6	ENSG00000204839	642475	-0.54688621687734	4.24969657797877	-4.14679600171064	0.0012375061261235	0.241003896378655	-0.788366548851609
ENSG00000123095	BHLHE41	ENSG00000123095	79365	0.587414517765713	4.48210799464343	4.1168719196503	0.00130721805700425	0.241003896378655	-0.813453718469556
ENSG00000113742	CPEB4	ENSG00000113742	80315	0.740629098422608	4.23532472239876	4.09670393690343	0.00135647543839487	0.241003896378655	-0.870736996703509
ENSG00000153094	BCL2L1	ENSG00000153094	10018	0.509069577140568	5.14339605159028	3.87445447217029	0.00204451121671479	0.241003896378655	-1.15316283041863
ENSG00000144824	PHLDB2	ENSG00000144824	90102	0.511554638264126	4.58731628263346	3.83826911673392	0.00218668787207618	0.241003896378655	-1.23051019284478
ENSG00000142871	CCN1	ENSG00000142871	3491	1.99704684284554	0.194576032986446	3.79380940138015	0.00237534402420724	0.241003896378655	-2.71624354141695
ENSG00000078081	LAMP3	ENSG00000078081	27074	1.34931038762069	1.76564696583003	3.68066672859388	0.00293423308793818	0.241003896378655	-2.09453554814323
ENSG00000139112	GABARAPL1	ENSG00000139112	23710	0.676136563787029	4.55515319202091	3.6758729955669	0.00296068287000238	0.241003896378655	-1.48569050494023
ENSG00000146411	SLC2A12	ENSG00000146411	154091	0.545751758652775	4.53569629296539	3.66446485710779	0.00302461073334117	0.241003896378655	-1.50344387331073
ENSG00000019485	PRDM11	ENSG00000019485	56981	0.604905594031952	1.20760374210462	3.65063919050334	0.00310397642248564	0.241003896378655	-2.31175759565628
ENSG00000104081	BMF	ENSG00000104081	90427	0.591755808335062	2.40474779096709	3.60859688849318	0.0033586104092759	0.241003896378655	-1.91103176335816
ENSG00000105290	APLP1	ENSG00000105290	333	0.987155033622569	0.193217041850965	3.60677481163843	0.00337011526964431	0.241003896378655	-2.78203194032882
ENSG00000105974	CAV1	ENSG00000105974	857	1.29012905386002	1.19403040902238	3.56719373512143	0.00363018401920429	0.241003896378655	-2.42522968649896
ENSG00000120885	CLU	ENSG00000120885	1191	1.24976408832206	4.94029517388274	3.55683787566185	0.00370153757981514	0.241003896378655	-1.66538413627945
ENSG00000184205	TSPYL2	ENSG00000184205	64061	1.450378189186	3.96923769442069	3.5488772841492	0.00375735556859227	0.241003896378655	-1.72771170897396
ENSG00000162645	GBP2	ENSG00000162645	2634	0.748890751490964	3.72256377709576	3.50586957229467	0.00407405538417303	0.241003896378655	-1.80059464550431
ENSG00000128590	DNAJB9	ENSG00000128590	4189	1.0301961333564	4.20037526555525	3.4831073709035	0.00425250877303622	0.241003896378655	-1.80380030295071
ENSG00000151117	TMEM86A	ENSG00000151117	144110	0.631453231071928	1.53316221785292	3.43647058061468	0.00464330272309873	0.241003896378655	-2.3972948909283

02. SW948/RKO 方案中的 RKO DEG 子集

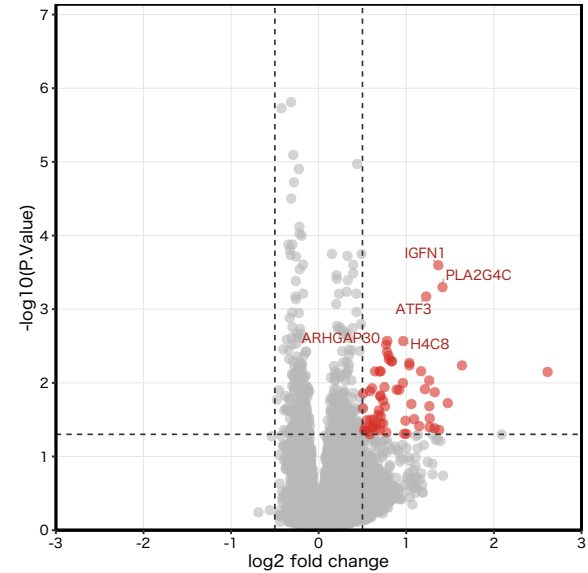
说明：该表将交集基因回填到对应差异分析结果中，展示同一候选基因在单个细胞系 LRC vs BULK 比较中的方向、效应量和显著性。

相对路径：`results/ngs/GSE114012/intersect/SW948_RKO/RKO/deg_results.csv`

表格预览：展示全部列与前 21 行。

GeneID	Symbol	Ensembl	Entrez	logFC	AveExpr	t	P.Value	adj.P.Val	B
ENSG00000187098	MITF	ENSG00000187098	4286	0.8111024559283	2.18602023225547	4.70757418824593	0.000257052070654833	0.00160435031384931	0.339637806391534
ENSG00000165029	ABCA1	ENSG00000165029	19	0.553988547972144	0.582013409020213	2.13884296409381	0.0487314032220465	0.0938194096465256	-4.36991902168681
ENSG00000118503	TNFAIP3	ENSG00000118503	7128	0.555741845760884	1.35597953453953	2.95002113969844	0.00965618289670823	0.0255570415909439	-3.04065183157421
ENSG00000204839	MROH6	ENSG00000204839	642475	0.524398071158957	1.10170369330259	2.61336827082563	0.0191756172992483	0.0441869738834234	-3.63193202962416
ENSG00000123095	BHLHE41	ENSG00000123095	79365	1.84897096912068	2.30270698332336	9.16525849925007	1.17907904482143e-07	8.2049859858661e-06	8.01771873673729
ENSG00000113742	CPEB4	ENSG00000113742	80315	0.808639532109899	4.14659221587045	8.33704197020787	4.05344660661201e-07	1.8334634707136e-05	6.56188229631066
ENSG00000153094	BCL2L1	ENSG00000153094	10018	0.806790089433523	4.36707502793424	7.80450807244418	9.36733172380214e-07	3.14167386536375e-05	5.66935220618103
ENSG00000144824	PHLDB2	ENSG00000144824	90102	0.694571395918303	0.824464267697845	3.04129912862232	0.00799975330800174	0.022072885698135	-2.74827215149362
ENSG00000142871	CCN1	ENSG00000142871	3491	-0.761772801020671	8.64522807943476	-10.7380361065464	1.38728311416259e-08	2.06842199626439e-06	9.84507154163788
ENSG00000078081	LAMP3	ENSG00000078081	27074	0.558799262370128	1.57464171316062	3.11257533509561	0.00690333377930467	0.0196770266822427	-2.76997065296182
ENSG00000139112	GABARAPL1	ENSG00000139112	23710	0.719586627760778	3.49664441811787	6.12336329207835	1.68296211073204e-05	0.000219466369391823	2.84143002670609
ENSG00000146411	SLC2A12	ENSG00000146411	154091	0.99475805376182	2.36841748712735	6.51581290123258	8.28564059074605e-06	0.000133777648949786	3.76106124486743
ENSG00000019485	PRDM11	ENSG00000019485	56981	0.719677811267038	1.81910300437341	3.59606351131245	0.00252564008815297	0.00895804855936078	-1.84537138483316
ENSG00000104081	BMF	ENSG00000104081	90427	1.41477124661262	0.973454650424807	5.88993750679928	2.59100142323698e-05	0.000298734214999464	2.79258805484107
ENSG00000105290	APLP1	ENSG00000105290	333	0.720966391106338	5.17272583191736	5.60625713500186	4.42093123321783e-05	0.000435849685743442	1.58784786995618
ENSG00000105974	CAV1	ENSG00000105974	857	1.41915878091584	4.96628547381708	17.4113269250829	1.34552480067727e-11	1.72376883876672e-08	16.9843367170902
ENSG00000120885	CLU	ENSG00000120885	1191	0.853606018765091	7.70161254116287	9.46913358096589	7.64321173117204e-08	6.1951052400916e-06	8.04690501964738
ENSG00000184205	TSPYL2	ENSG00000184205	64061	1.48635387200305	5.88792403696203	14.4629463911397	2.04135959931767e-10	1.17397618480759e-07	14.212877472322
ENSG00000162645	GBP2	ENSG00000162645	2634	2.66935736541961	1.03608666484377	10.4575034089105	1.99554918999382e-08	2.67780528528393e-06	9.48084914723546
ENSG00000128590	DNAJB9	ENSG00000128590	4189	0.517682678448129	4.56001669638491	5.32754578498834	7.55073157860792e-05	0.000648249175282002	1.1223445716708
ENSG00000151117	TMEM86A	ENSG00000151117	144110	0.622228979664469	1.94887153122282	3.78829803103482	0.00169285766118795	0.00659558274155064	-1.48490209478124

03. 传统火山图：ALL



说明：

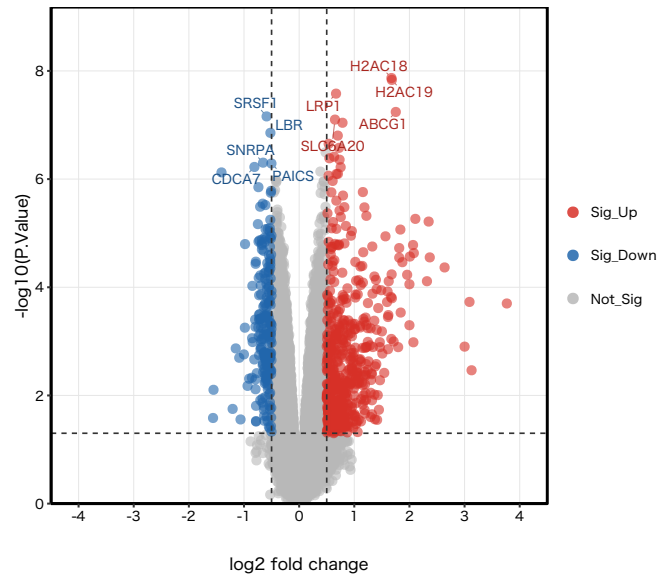
- 火山图展示 LRC 相对 BULK 的效应量与显著性分布，红/蓝分别提示休眠样状态中的上调和下调信号；
- 标注基因可作为后续 TF 调控、通路富集和药物逆转 signature 的候选入口；
- 单组图用于判断该模型中休眠-增殖状态切换的主要表达漂移方向。

相对路径：

`results/ngs/GSE114012/plots/volcano/ALL/volcano_plot.pdf`

- Sig_Up
- Not_Sig

03. 传统火山图：DLD1



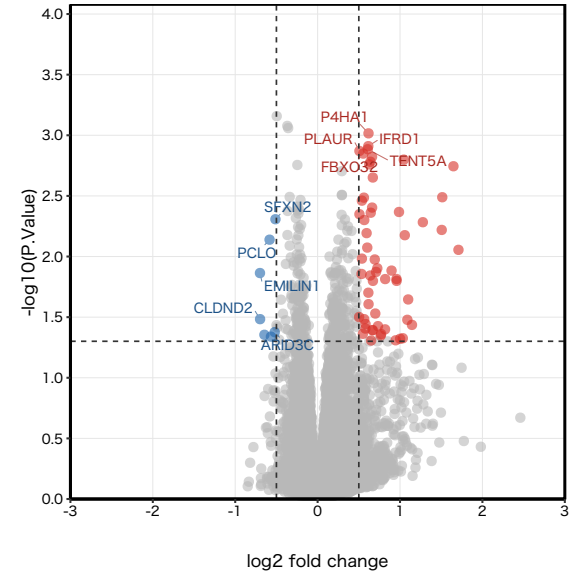
说明：

- 火山图展示 LRC 相对 BULK 的效应量与显著性分布，红/蓝分别提示休眠样状态中的上调和下调信号；
- 标注基因可作为后续 TF 调控、通路富集和药物逆转 signature 的候选入口；
- 单组图用于判断该模型中休眠-增殖状态切换的主要表达漂移方向。

相对路径：

`results/ngs/GSE114012/plots/volcano/DLD1/volcano_plot.pdf`

03. 传统火山图：DLD1/HCT15



说明：

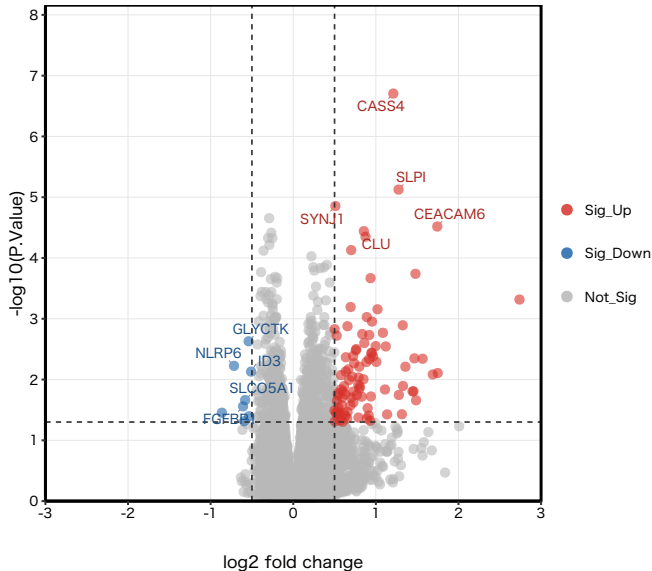
- 火山图展示 LRC 相对 BULK 的效应量与显著性分布，红/蓝分别提示休眠样状态中的上调和下调信号；
- 标注基因可作为后续 TF 调控、通路富集和药物逆转 signature 的候选入口；
- 单组图用于判断该模型中休眠-增殖状态切换的主要表达漂移方向。

相对路径：

results/ngs/GSE114012/plots/volcano/DLD1_HCT15/volcano_plot.pdf

- Sig_Up
- Sig_Down
- Not_Sig

03. 传统火山图：DLD1/HCT15/SW48



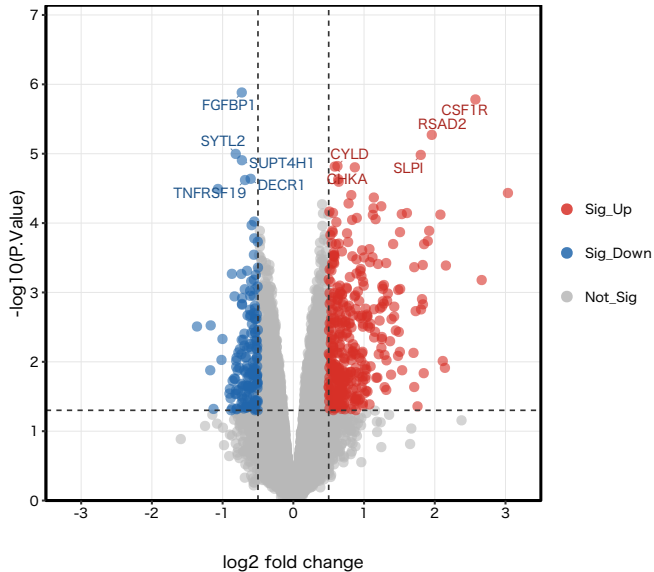
说明：

- 火山图展示 LRC 相对 BULK 的效应量与显著性分布，红/蓝分别提示休眠样状态中的上调和下调信号；
- 标注基因可作为后续 TF 调控、通路富集和药物逆转 signature 的候选入口；
- 单组图用于判断该模型中休眠-增殖状态切换的主要表达漂移方向。

相对路径：

`results/ngs/GSE114012/plots/volcano/DLD1_HCT15_SW48/volcano_plot.pdf`

03. 传统火山图：HCT15



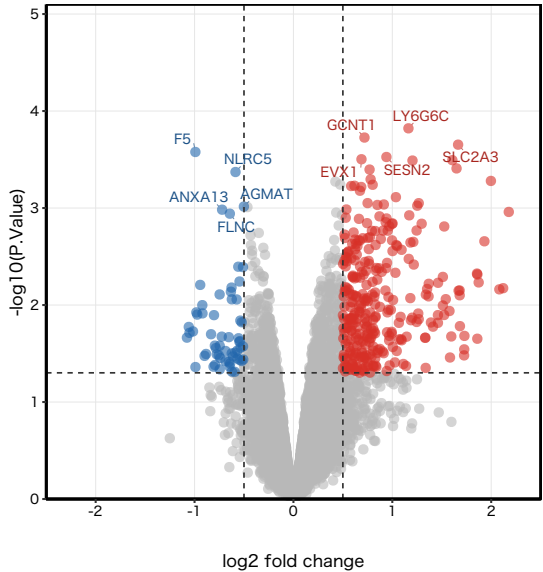
说明：

- 火山图展示 LRC 相对 BULK 的效应量与显著性分布，红/蓝分别提示休眠样状态中的上调和下调信号；
- 标注基因可作为后续 TF 调控、通路富集和药物逆转 signature 的候选入口；
- 单组图用于判断该模型中休眠-增殖状态切换的主要表达漂移方向。

相对路径：

results/ngs/GSE114012/plots/volcano/HCT15/volcano_plot.pdf

03. 传统火山图：HT55



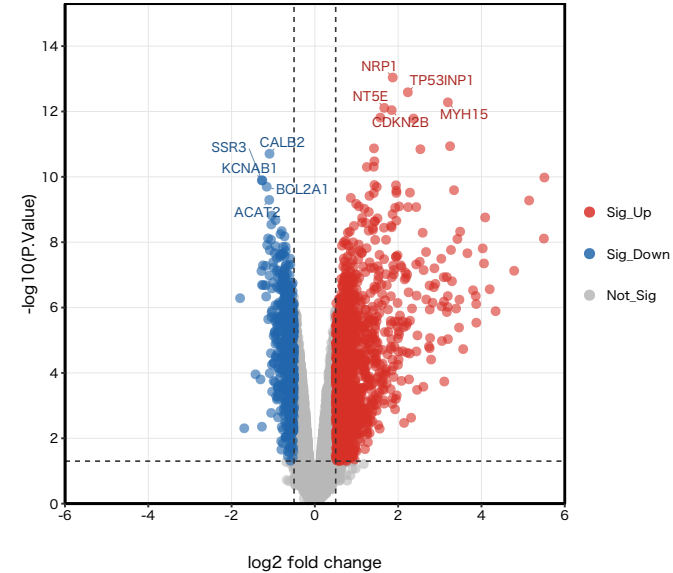
说明：

- 火山图展示 LRC 相对 BULK 的效应量与显著性分布，红/蓝分别提示休眠样状态中的上调和下调信号；
- 标注基因可作为后续 TF 调控、通路富集和药物逆转 signature 的候选入口；
- 单组图用于判断该模型中休眠-增殖状态切换的主要表达漂移方向。

相对路径：

`results/ngs/GSE114012/plots/volcano/HT55/volcano_plot.pdf`

03. 传统火山图：RKO



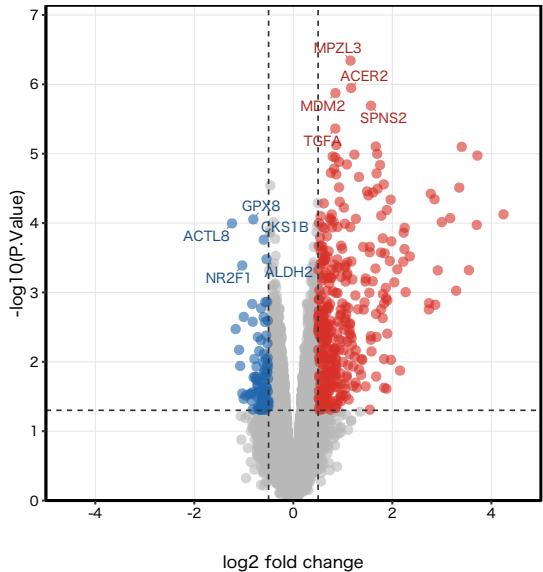
说明：

- 火山图展示 LRC 相对 BULK 的效应量与显著性分布，红/蓝分别提示休眠样状态中的上调和下调信号；
- 标注基因可作为后续 TF 调控、通路富集和药物逆转 signature 的候选入口；
- 单组图用于判断该模型中休眠-增殖状态切换的主要表达漂移方向。

相对路径：

`results/ngs/GSE114012/plots/volcano/RKO/volcano_plot.pdf`

03. 传统火山图：SW48



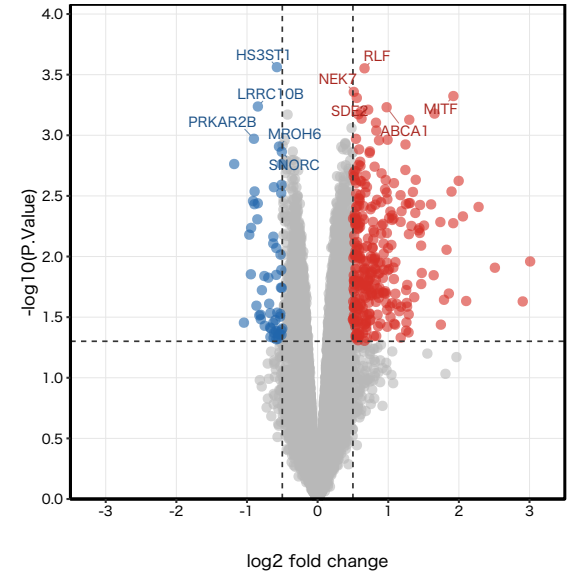
说明：

- 火山图展示 LRC 相对 BULK 的效应量与显著性分布，红/蓝分别提示休眠样状态中的上调和下调信号；
- 标注基因可作为后续 TF 调控、通路富集和药物逆转 signature 的候选入口；
- 单组图用于判断该模型中休眠-增殖状态切换的主要表达漂移方向。

相对路径：

`results/ngs/GSE114012/plots/volcano/SW48/volcano_plot.pdf`

03. 传统火山图：SW948



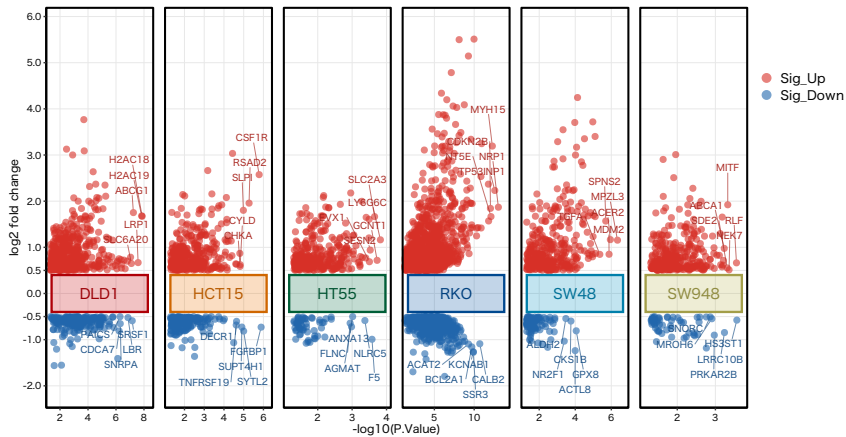
说明：

- 火山图展示 LRC 相对 BULK 的效应量与显著性分布，红/蓝分别提示休眠样状态中的上调和下调信号；
- 标注基因可作为后续 TF 调控、通路富集和药物逆转 signature 的候选入口；
- 单组图用于判断该模型中休眠-增殖状态切换的主要表达漂移方向。

相对路径：

results/ngs/GSE114012/plots/volcano/SW948/volcano_plot.pdf

04. 多组火山图： ALL



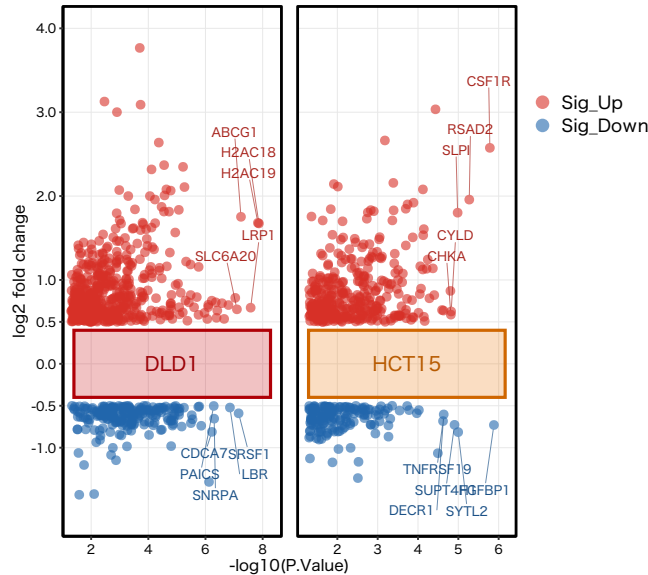
说明：

- 多组火山图并列展示多个细胞系或组合方案中的显著基因分布，便于观察跨模型方向一致性；
- 稳定重复出现的上调/下调信号更适合作为休眠维持或复苏触发轴的候选基因；
- 该图为从单模型差异走向共同调控网络提供直观筛选依据。

相对路径：

results/ngs/GSE114012/plots/multiple_volcano/ALL/multiple_volcano_plot.pdf

04. 多组火山图：DLD1/HCT15



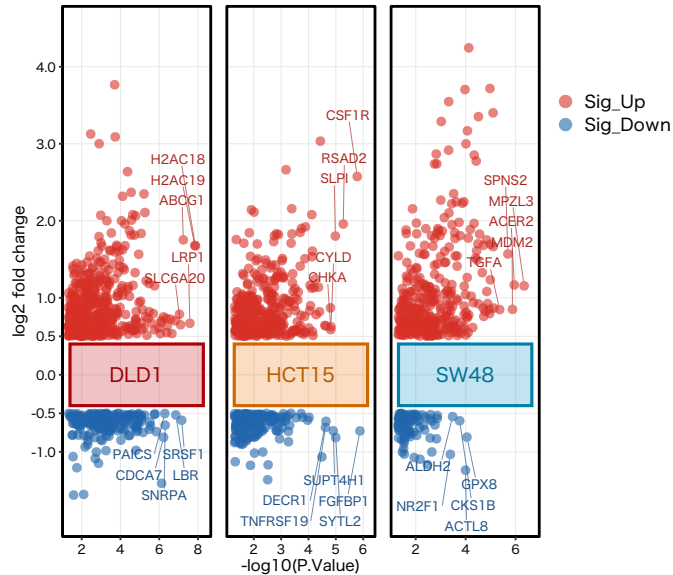
说明：

- 多组火山图并列展示多个细胞系或组合方案中的显著基因分布，便于观察跨模型方向一致性；
- 稳定重复出现的上调/下调信号更适合作为休眠维持或复苏触发轴的候选基因；
- 该图为从单模型差异走向共同调控网络提供直观筛选依据。

相对路径：

results/ngs/GSE114012/plots/multiple_volcano/DLD1_HCT15/
multiple_volcano_plot.pdf

04. 多组火山图：DLD1/HCT15/SW48



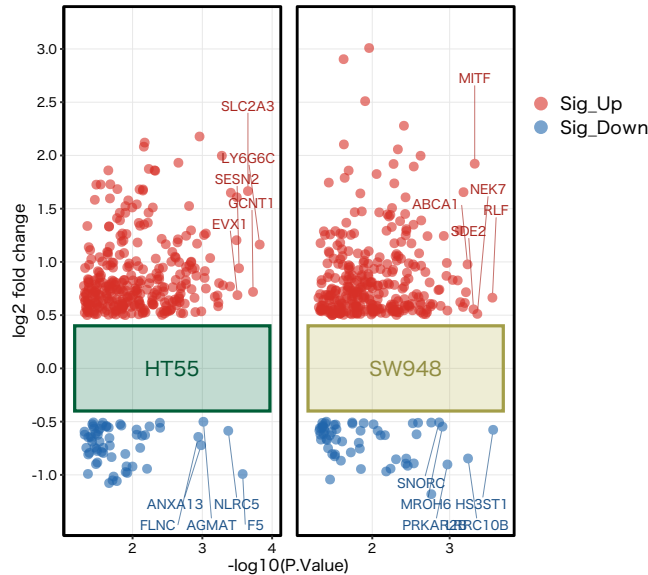
说明：

- 多组火山图并列展示多个细胞系或组合方案中的显著基因分布，便于观察跨模型方向一致性；
- 稳定重复出现的上调/下调信号更适合作为休眠维持或复苏触发轴的候选基因；
- 该图为从单模型差异走向共同调控网络提供直观筛选依据。

相对路径：

results/ngs/GSE114012/plots/multiple_volcano/DLD1_HCT15_SW48/multiple_volcano_plot.pdf

04. 多组火山图：HT55/SW948



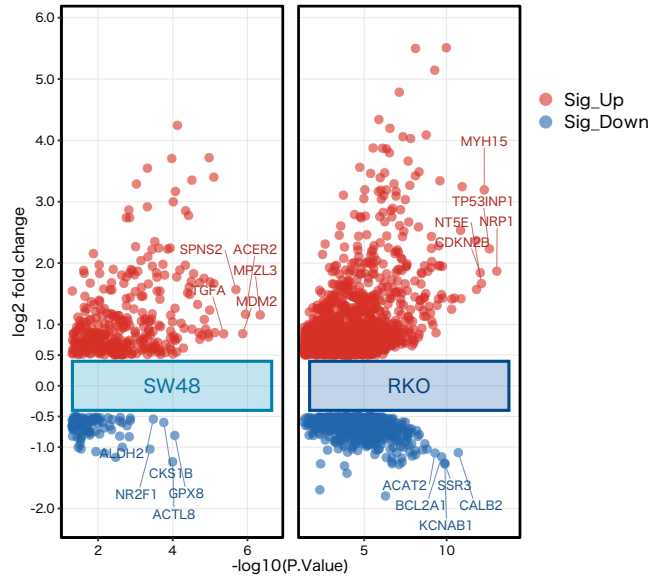
说明：

- 多组火山图并列展示多个细胞系或组合方案中的显著基因分布，便于观察跨模型方向一致性；
- 稳定重复出现的上调/下调信号更适合作为休眠维持或复苏触发轴的候选基因；
- 该图为从单模型差异走向共同调控网络提供直观筛选依据。

相对路径：

results/ngs/GSE114012/plots/multiple_volcano/HT55_SW948/
multiple_volcano_plot.pdf

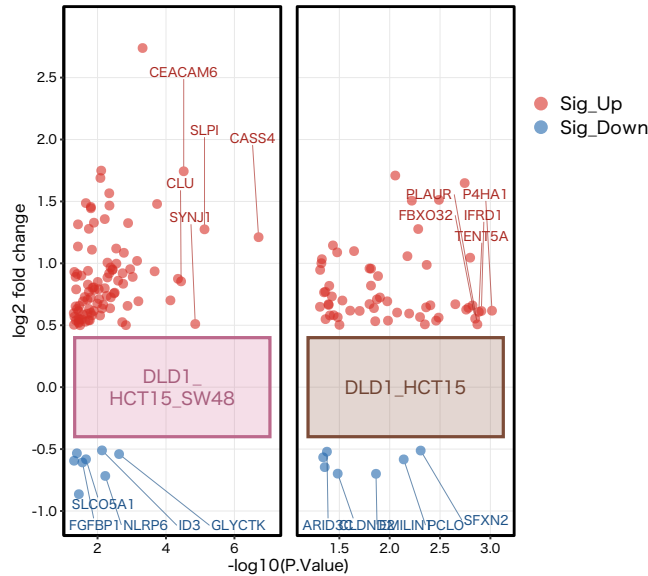
04. 多组火山图：SW48/RKO



- 说明：
- 多组火山图并列展示多个细胞系或组合方案中的显著基因分布，便于观察跨模型方向一致性；
 - 稳定重复出现的上调/下调信号更适合作为休眠维持或复苏触发轴的候选基因；
 - 该图为从单模型差异走向共同调控网络提供直观筛选依据。

相对路径：
results/ngs/GSE114012/plots/multiple_volcano/SW48_RKO/
multiple_volcano_plot.pdf

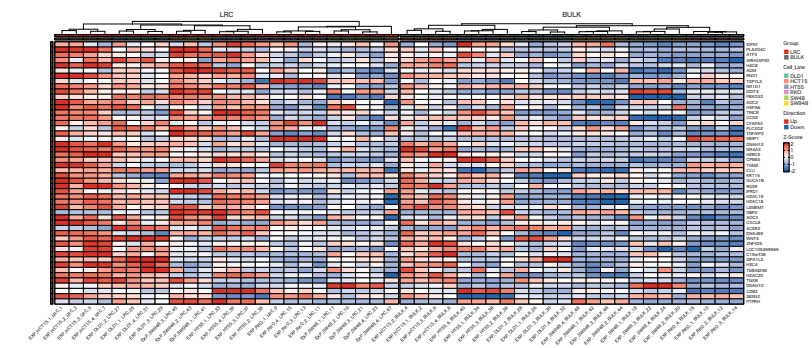
04. 多组火山图： COMBINATION 1



- 说明：
- 多组火山图并列展示多个细胞系或组合方案中的显著基因分布，便于观察跨模型方向一致性；
 - 稳定重复出现的上调/下调信号更适合作为休眠维持或复苏触发轴的候选基因；
 - 该图为从单模型差异走向共同调控网络提供直观筛选依据。

相对路径：
`results/ngs/GSE114012/plots/multiple_volcano/COMBINATION_1/multiple_volcano_plot.pdf`

05. Top 50 差异基因热图：ALL



说明：

- 该图基于显著差异基因表按 P 值与 logFC 幅度筛选 Top 50 基因，并回到 TPM 矩阵中计算 $\log_2(TPM+1)$ 行 Z-Score；
- LRC 统一置于左侧、BULK 置于右侧，便于直接判断休眠样状态与相对增殖状态之间的表达分离；
- 红蓝方向条带提示候选基因在 LRC 中的上调或下调方向，可作为后续休眠维持/复苏候选轴筛选依据。

相对路径：

results/ngs/GSE114012/plots/gene_heatmap/ALL/gene_heatmap.pdf

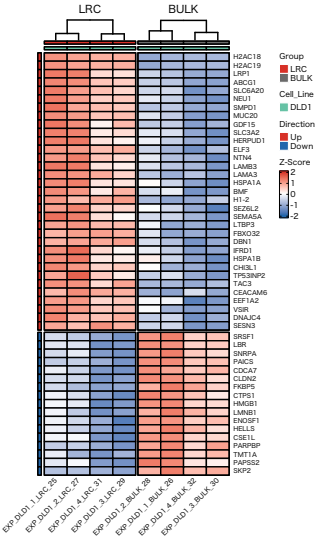
05. Top 50 差异基因热图：DLD1

说明：

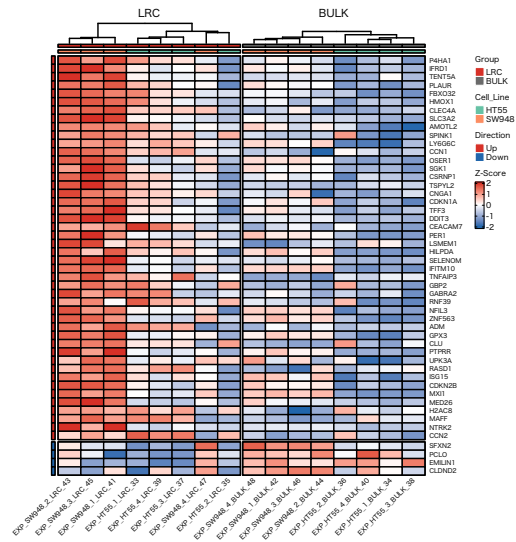
- 该图基于显著差异基因表按 P 值与 logFC 幅度筛选 Top 50 基因，并回到 TPM 矩阵中计算 $\log_2(\text{TPM}+1)$ 行 Z-Score；
- LRC 统一置于左侧、BULK 置于右侧，便于直接判断休眠样状态与相对增殖状态之间的表达分离；
- 红蓝方向条带提示候选基因在 LRC 中的上调或下调方向，可作为后续休眠维持/复苏候选轴筛选依据。

相对路径：

results/ngs/GSE114012/plots/gene_heatmap/DLD1/gene_heatmap.pdf



05. Top 50 差异基因热图：DLD1/HCT15



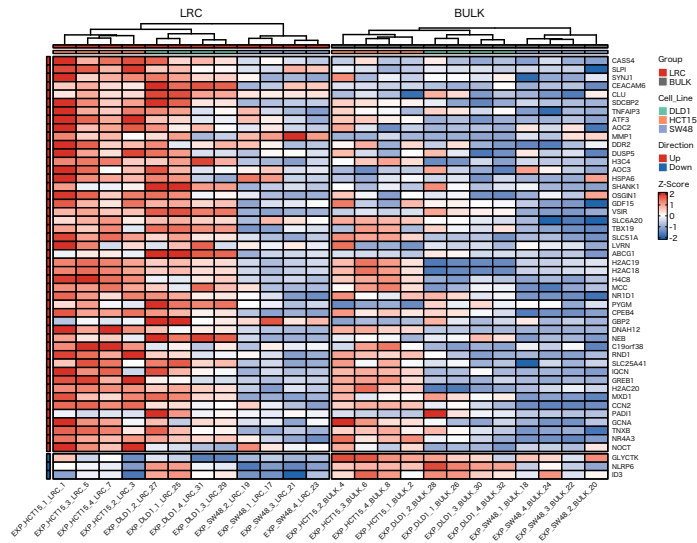
说明：

- 该图基于显著差异基因表按 P 值与 logFC 幅度筛选 Top 50 基因，并回到 TPM 矩阵中计算 $\log_2(\text{TPM}+1)$ 行 Z-Score；
- LRC 统一置于左侧、BULK 置于右侧，便于直接判断休眠样状态与相对增殖状态之间的表达分离；
- 跨细胞系组合更强调共同表达模式，有助于从单细胞系特异信号中提炼更稳健的休眠相关候选基因。

相对路径：

results/ngs/GSE114012/plots/gene_heatmap/DLD1_HCT15/gene_heatmap.pdf

05. Top 50 差异基因热图：DLD1/HCT15/SW48



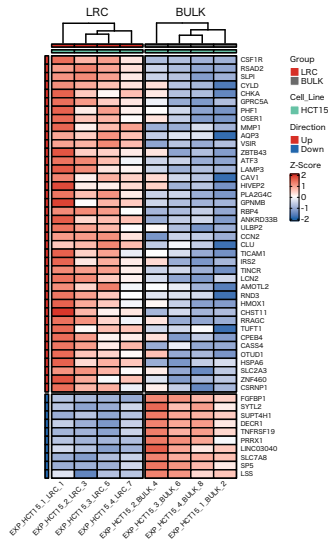
说明：

- 该图基于显著差异基因表按 P 值与 logFC 幅度筛选 Top 50 基因，并回到 TPM 矩阵中计算 $\log_2(\text{TPM}+1)$ 行 Z-Score；
- LRC 统一置于左侧、BULK 置于右侧，便于直接判断休眠样状态与相对增殖状态之间的表达分离；
- 该组合可用于观察多模型共同休眠样表达特征，为后续 TF 调控、通路富集和药物重定位提供更严格入口。

相对路径：

`results/ngs/GSE114012/plots/gene_heatmap/
DLD1_HCT15_SW48/gene_heatmap.pdf`

05. Top 50 差异基因热图：HCT15



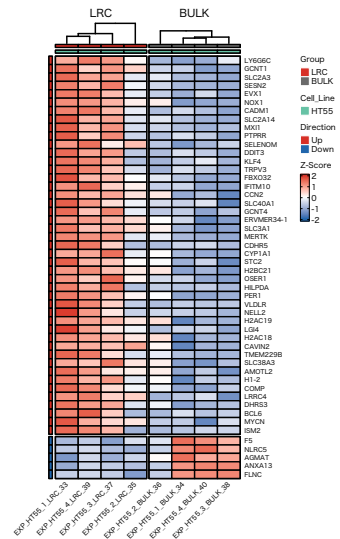
说明：

- 该图基于显著差异基因表按 P 值与 logFC 幅度筛选 Top 50 基因，并回到 TPM 矩阵中计算 $\log_2(\text{TPM}+1)$ 行 Z-Score；
- LRC 统一置于左侧、BULK 置于右侧，便于直接判断休眠样状态与相对增殖状态之间的表达分离；
- 单细胞系热图可检查该模型自身的 LRC/BULK 表达分界，并辅助识别与组合分析一致的候选基因。

相对路径：

results/ngs/GSE114012/plots/gene_heatmap/HCT15/gene_heatmap.pdf

05. Top 50 差异基因热图：HT55



说明：

- 该图基于显著差异基因表按 P 值与 logFC 幅度筛选 Top 50 基因，并回到 TPM 矩阵中计算 $\log_2(\text{TPM}+1)$ 行 Z-Score；
- LRC 统一置于左侧、BULK 置于右侧，便于直接判断休眠样状态与相对增殖状态之间的表达分离；
- 单细胞系热图可检查该模型自身的 LRC/BULK 表达分界，并辅助识别与组合分析一致的候选基因。

相对路径：

results/ngs/GSE114012/plots/gene_heatmap/HT55/gene_heatmap.pdf

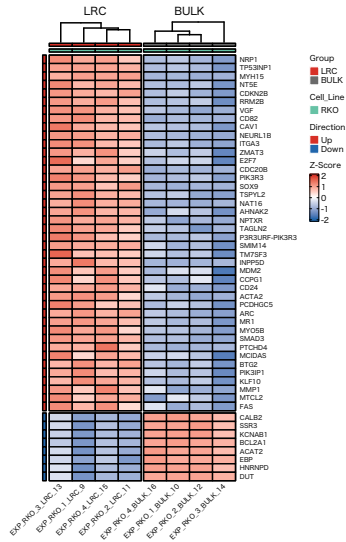
05. Top 50 差异基因热图：RKO

说明：

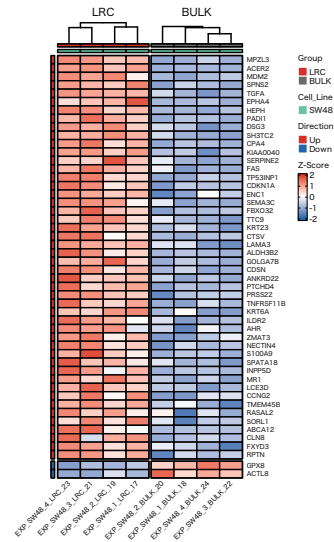
- 该图基于显著差异基因表按 P 值与 logFC 幅度筛选 Top 50 基因，并回到 TPM 矩阵中计算 $\log_2(\text{TPM}+1)$ 行 Z-Score；
- LRC 统一置于左侧、BULK 置于右侧，便于直接判断休眠样状态与相对增殖状态之间的表达分离；
- 单细胞系热图可检查该模型自身的 LRC/BULK 表达分界，并辅助识别与组合分析一致的候选基因。

相对路径：

results/ngs/GSE114012/plots/gene_heatmap/RKO/gene_heatmap.pdf



05. Top 50 差异基因热图：SW48



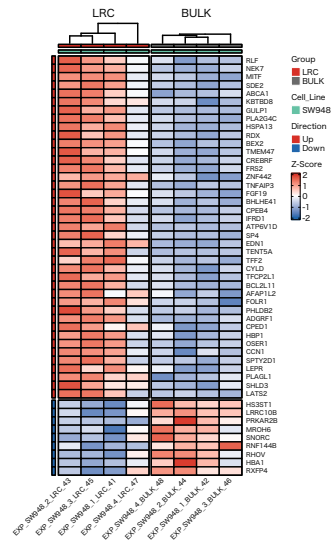
说明：

- 该图基于显著差异基因表按 P 值与 logFC 幅度筛选 Top 50 基因，并回到 TPM 矩阵中计算 $\log_2(\text{TPM}+1)$ 行 Z-Score；
- LRC 统一置于左侧、BULK 置于右侧，便于直接判断休眠样状态与相对增殖状态之间的表达分离；
- 单细胞系热图可检查该模型自身的 LRC/BULK 表达分界，并辅助识别与组合分析一致的候选基因。

相对路径：

results/ngs/GSE114012/plots/gene_heatmap/SW48/gene_heatmap.pdf

05. Top 50 差异基因热图：SW948



说明：

- 该图基于显著差异基因表按 P 值与 logFC 幅度筛选 Top 50 基因，并回到 TPM 矩阵中计算 $\log_2(\text{TPM}+1)$ 行 Z-Score；
- LRC 统一置于左侧、BULK 置于右侧，便于直接判断休眠样状态与相对增殖状态之间的表达分离；
- 单细胞系热图可检查该模型自身的 LRC/BULK 表达分界，并辅助识别与组合分析一致的候选基因。

相对路径：

results/ngs/GSE114012/plots/gene_heatmap/SW948/gene_heatmap.pdf